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In [ ]: # =====
# CELL 1: IMPORTS & SETUP
# =====

import warnings
warnings.filterwarnings('ignore')

import pytensor
pytensor.config.mode = 'FAST_COMPILE'

import pandas as pd
import numpy as np
import pymc as pm
import arviz as az
import unicodedata
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12, 6)
plt.rcParams['font.size'] = 10
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In [ ]: # =====
# CELL 2: CONSTANTS & HELPER FUNCTIONS
# =====

VALID_LEAGUES = ['Premier League', 'Ligue 1', 'Bundesliga', 'Serie A', 'La Liga']

LEAGUE_MAP = {
    'Ligue 1': 0,
    'Bundesliga': 1,
    'Serie A': 2,
    'La Liga': 3,
    'Premier League': 4,
}

LEAGUE_MAP_REVERSE = {v: k for k, v in LEAGUE_MAP.items()}

COMMON_STATS_ATT_MID = ['KP', 'PrgP', 'PrgC', 'SCA']
COMMON_STATS_DEF = ['TklW', 'Int', 'Clr', 'Blocks']

POSITION_FILTERS_ATT_MID = {
    'ATT': ['FWD'],
    'MID': ['MID'],
}

POSITION_FILTERS_DEF = {
    'DEF': ['DEF'],
}

# 2022/23→2023/24 & 2023/24→2024/25 is the held-out OOS test – never enters training
EXCLUDED_TRANSFER_SEASONS = {
    '2022_2023_to_2023_2024',
    '2023_2024_to_2024_2025',
}
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}

def normalize_name(x: str) -> str:
    x = unicodedata.normalize('NFKD', str(x))
    x = x.encode('ascii', 'ignore').decode('ascii')
    return x.replace(' ', '').lower()

def bucket_position(pos: str) -> str:
    primary = str(pos).upper().split(',')[0].strip()
    if 'D' in primary:
        return 'DEF'
    elif 'M' in primary:
        return 'MID'
    elif 'F' in primary:
        return 'FWD'
    return 'OTHER'

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In [ ]: # =====
# CELL 3: DATA LOADING FUNCTIONS
# =====

def _standardize_league_and_position(df: pd.DataFrame) -> pd.DataFrame:
    df['Comp'] = (
        df['Comp']
        .astype(str)
        .str.replace(r'^[a-z]{2,3}\s+', '', regex=True)
        .str.strip()
    )
    df['Comp'] = df['Comp'].replace({'eng Premier League': 'Premier League'})
    df['Pos_Bucket'] = df['Pos'].apply(bucket_position)
    return df

def _load_base_historical(path: str) -> pd.DataFrame:
    """Load a historical season CSV (2017/18-2023/24).
    NOTE: 'Avg Mins per Match' is actually total season minutes (column name is
    misleading). 90s = total_minutes / 90. Do NOT multiply by Matches Played.
    """
    df = pd.read_csv(path)
    df.columns = [c.strip() for c in df.columns]
    df = df.rename(columns={
        'player': 'Player',
        'squad': 'Squad',
        'comp': 'Comp',
        'pos': 'Pos',
        'age': 'Age',
    })
    # FIX: 'Avg Mins per Match' is total season minutes, not average per game.
    # Previous code erroneously multiplied by 'Matches Played', inflating 90s ~20x.
    df['90s'] = pd.to_numeric(df['Avg Mins per Match'], errors='coerce').fillna(0) / 90.0
    return df

def _load_base_modern(path: str) -> pd.DataFrame:
    """Load a modern season CSV (2024/25+): title-case cols, '90s' already present."""
    df = pd.read_csv(path)

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df.columns = [c.strip() for c in df.columns]
if '90s' not in df.columns and 'Min' in df.columns:
    df['90s'] = pd.to_numeric(df['Min'], errors='coerce').fillna(0) / 90.0
return df

def load_season_att_mid(path: str, is_historical: bool = True, stats_list=None) -> pd.DataFrame
    if is_historical:
        df = _load_base_historical(path)
        df['KP'] = pd.to_numeric(df['Key passes'], errors='coerce').fillna(0)
        df['PrgP'] = pd.to_numeric(df['Progressive Passes'], errors='coerce').fillna(0)
        df['PrgC'] = pd.to_numeric(df['Progressive Carries'], errors='coerce').fillna(0)
        df['SCA'] = (
            pd.to_numeric(df['Shot creating actions p 90'], errors='coerce').fillna(0)
            * df['90s']
        )
    else:
        df = _load_base_modern(path)
        if 'SCA' not in df.columns and 'SCA90' in df.columns:
            df['SCA'] = pd.to_numeric(df['SCA90'], errors='coerce').fillna(0) * df['90s']
        elif 'SCA' in df.columns:
            df['SCA'] = pd.to_numeric(df['SCA'], errors='coerce').fillna(0)
        else:
            df['SCA'] = 0.0
        for col in ['KP', 'PrgP', 'PrgC']:
            if col in df.columns:
                df[col] = pd.to_numeric(df[col], errors='coerce').fillna(0)
    if stats_list is None:
        stats_list = COMMON_STATS_ATT_MID
    for stat in stats_list:
        if stat not in df.columns:
            df[stat] = 0.0
        else:
            df[stat] = pd.to_numeric(df[stat], errors='coerce').fillna(0)
    df = _standardize_league_and_position(df)
    df = df[df['Comp'].isin(VALID_LEAGUES)]
    df = df[df['90s'] >= 3.0]
    df['Player_Key'] = df['Player'].apply(normalize_name)
    return df.reset_index(drop=True)

def load_season_def(path: str, is_historical: bool = True, stats_list=None) -> pd.DataFrame
    if is_historical:
        df = _load_base_historical(path)
        df = df.rename(columns={
            'Tackles Won': 'TklW',
            'Interceptions': 'Int',
            'Clearances': 'Clr',
        })
        shots_blk = pd.to_numeric(df['Shots blocked'], errors='coerce').fillna(0)
        passes_blk = pd.to_numeric(df['Passes blocked'], errors='coerce').fillna(0)
        df['Blocks'] = shots_blk + passes_blk
    else:
        df = _load_base_modern(path)
        for col in ['TklW', 'Int', 'Clr']:
            if col in df.columns:
                df[col] = pd.to_numeric(df[col], errors='coerce').fillna(0)
        # FIX: 'Blocks' in the main table is shots blocked only.

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# 'Blocks_stats_defense' = Sh_stats_defense + Pass, which exactly
# matches the historical definition of Shots blocked + Passes blocked.
if 'Blocks_stats_defense' in df.columns:
    df['Blocks'] = pd.to_numeric(df['Blocks_stats_defense'], errors='coerce').fillna(0)
elif 'Blocks' in df.columns:
    df['Blocks'] = pd.to_numeric(df['Blocks'], errors='coerce').fillna(0)
else:
    df['Blocks'] = 0.0
if stats_list is None:
    stats_list = COMMON_STATS_DEF
for stat in stats_list:
    if stat not in df.columns:
        df[stat] = 0.0
    else:
        df[stat] = pd.to_numeric(df[stat], errors='coerce').fillna(0)
df = _standardize_league_and_position(df)
df = df[df['Pos_Bucket'] == 'DEF'].copy()
df = df[df['Comp'].isin(VALID_LEAGUES)]
df = df[df['90s'] >= 3.0]
df['Player_Key'] = df['Player'].apply(normalize_name)
return df.reset_index(drop=True)

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In [ ]: # =====
# CELL 4: BRIDGE CONSTRUCTION FUNCTION
# =====

def build_bridge(
    data_dict: dict,
    valid_positions: list,
    stats_list: list,
    train_seasons: list,
    label: str,
):
    """
    Build the training bridge DataFrame from consecutive season pairs.

    New features vs previous version:
    - League_origin : original league name string (not just numeric ID)
    - Transfer_Season: season-pair label e.g. '2017_2018_to_2018_2019'
    - Excluded pairs : any pair touching EXCLUDED_TRANSFER_SEASONS is skipped
    - Assertion      : all League_ID values must match LEAGUE_MAP before returning
    """
    rows = []

    for i in range(len(train_seasons) - 1):
        s0, s1 = train_seasons[i], train_seasons[i + 1]
        transfer_season = f'{s0}_to_{s1}'

        # Hard temporal truncation - never let held-out season pairs leak in
        if transfer_season in EXCLUDED_TRANSFER_SEASONS:
            print(f" Skipping excluded pair: {transfer_season}")
            continue

        if s0 not in data_dict or s1 not in data_dict:
            print(f" WARNING: missing season data for pair {s0}/{s1}, skipping.")
            continue

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df_s0 = (
    data_dict[s0]
    .sort_values('90s', ascending=False)
    .drop_duplicates('Player_Key')
)
df_s1 = (
    data_dict[s1]
    .sort_values('90s', ascending=False)
    .drop_duplicates('Player_Key')
)

m = pd.merge(df_s0, df_s1, on='Player_Key', suffixes=('_src', '_trg'))

valid = m[
    (m['Squad_src'] != m['Squad_trg']) &
    (m['Comp_trg'] == 'Premier League') &
    (m['90s_src'] >= 5.0) &
    (m['90s_trg'] >= 5.0) &
    (m['Pos_Bucket_src'].isin(valid_positions))
].copy()

EPS = 0.05

for _, r in valid.iterrows():
    src_league = r['Comp_src']
    league_id = LEAGUE_MAP.get(src_league, -1)

    row = {
        'Player': r['Player_src'],
        'League_ID': league_id,
        'League_origin': src_league, # retain text label
        'Transfer_Season': transfer_season, # season-pair tag
        'is_foreign': int(src_league != 'Premier League'),
        'Age': r['Age_src'],
        'Position': r['Pos_Bucket_src'],
        'Team': r['Squad_trg'],
        '90s_src': r['90s_src'],
    }

    for s in stats_list:
        src_val = r.get(f'{s}_src', np.nan)
        trg_val = r.get(f'{s}_trg', np.nan)

        src_rate = src_val / r['90s_src']
        trg_rate = trg_val / r['90s_trg']

        row[f'{s}_src'] = src_rate
        row[f'{s}_trg'] = trg_rate
        row[f'{s}_log_ratio'] = np.log((trg_rate + EPS) / (src_rate + EPS))

    rows.append(row)

df_bridge = pd.DataFrame(rows).dropna(subset=['Age'])

if df_bridge.empty:

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        print(f"ERROR: No valid transfers found for {label}")
        return None, None, None, None, None

    for s in stats_list:
        col = f'{s}_log_ratio'
        if col in df_bridge.columns:
            df_bridge[col] = df_bridge[col].replace([np.inf, -np.inf], np.nan)

    log_ratio_cols = [f'{s}_log_ratio' for s in stats_list]
    df_bridge = df_bridge.dropna(subset=log_ratio_cols)

    if df_bridge.empty:
        print(f"ERROR: No valid transfers with clean log-ratios for {label}")
        return None, None, None, None, None

    # ---- Assertion: every League_ID must be a valid LEAGUE_MAP value ----
    valid_ids = set(LEAGUE_MAP.values())
    invalid_mask = ~df_bridge['League_ID'].isin(valid_ids)
    assert not invalid_mask.any(), (
        f"League_ID mapping mismatch in {label} bridge!\n"
        f"{df_bridge.loc[invalid_mask, ['Player', 'League_origin', 'League_ID']].head()}"
    )

    print(f" Bridge ({label}): {len(df_bridge)} transfers after cleaning")
    print(f" Transfer seasons: {sorted(df_bridge['Transfer_Season'].unique())}")

    age_mean = df_bridge['Age'].mean()
    df_bridge['Age_c'] = df_bridge['Age'] - age_mean

    df_bridge['team_id'], team_labels = pd.factorize(df_bridge['Team'])
    df_bridge['pos_id'], pos_labels = pd.factorize(df_bridge['Position'])
    df_bridge['player_id'], player_labels = pd.factorize(df_bridge['Player'])

    return df_bridge, age_mean, team_labels, pos_labels, log_ratio_cols

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In [ ]: # =====
# CELL 5: MODEL FITTING FUNCTION
# =====

def fit_joint_model(
    df_bridge: pd.DataFrame,
    stats_list: list,
    n_pos: int,
    n_teams: int,
):
    """
    Fit the joint multivariate hierarchical model.

    Prior change: league group variance now uses HalfStudentT(nu=4, sigma=0.5)
    instead of HalfNormal(0.5) to allow heavier tails and prevent over-shrinkage
    of defensive league effects toward zero.
    """
    n_stats = len(stats_list)

    standardization = {}
    Y_raw = np.zeros((len(df_bridge), n_stats))

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for j, s in enumerate(stats_list):
    col = f'{s}_log_ratio'
    vals = df_bridge[col].values
    mean = vals.mean()
    std = vals.std()
    if std < 1e-6:
        raise ValueError(
            f"Stat '{s}' has near-zero variance ({std:.2e}). "
            "Check data or remove this stat."
        )
    standardization[s] = {'mean': float(mean), 'std': float(std)}
    Y_raw[:, j] = (vals - mean) / std

Y = Y_raw

league_id = df_bridge['League_ID'].values.astype(int)
is_foreign = df_bridge['is_foreign'].values.astype(int)
pos_id = df_bridge['pos_id'].values.astype(int)
team_id = df_bridge['team_id'].values.astype(int)
age_c = (df_bridge['Age_c'].values / 5.0).astype(float)
minutes_src = df_bridge['90s_src'].values

w = np.sqrt(minutes_src / (minutes_src + 5.0))
Y_scaled = Y * w[:, None]

with pm.Model() as joint_model:
    alpha = pm.Normal('alpha', 0, 0.2, shape=n_stats)

    mu_lt = pm.Normal('mu_lt', 0, 0.2, shape=n_stats)
    # HalfStudentT: heavier tails than HalfNormal, prevents over-shrinkage
    sigma_lt = pm.HalfStudentT('sigma_lt', nu=4, sigma=0.5, shape=n_stats)
    z_league = pm.Normal('z_league', 0, 1, shape=(4, n_stats))
    league_tax = pm.Deterministic(
        'league_tax',
        mu_lt[None, :] + z_league * sigma_lt[None, :]
    )

    if n_pos > 1:
        sigma_pos = pm.HalfNormal('sigma_pos', 0.2)
        z_pos = pm.Normal('z_pos', 0, 1, shape=(n_pos, n_stats))
        pos_eff = pm.Deterministic('pos_eff', z_pos * sigma_pos)
    else:
        pos_eff = 0

    sigma_team = pm.HalfNormal('sigma_team', 0.2)
    z_team = pm.Normal('z_team', 0, 1, shape=(n_teams, n_stats))
    team_eff = pm.Deterministic('team_eff', z_team * sigma_team)

    beta_age = pm.Normal('beta_age', 0, 0.05, shape=n_stats)
    beta_age2 = pm.Normal('beta_age2', 0, 0.02, shape=n_stats)

    safe_league_idx = pm.intX(np.clip(league_id, 0, 3))
    pos_contribution = pos_eff[pos_id] if n_pos > 1 else 0

    mu = (

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        alpha
        + is_foreign[:, None] * league_tax[safe_league_idx]
        + pos_contribution
        + team_eff[team_id]
        + beta_age * age_c[:, None]
        + beta_age2 * (age_c[:, None] ** 2)
    )

    sd_dist = pm.HalfNormal.dist(0.3)
    chol, corr, sigma = pm.LKJCholeskyCov(
        'chol',
        n=n_stats,
        eta=3.0,
        sd_dist=sd_dist,
        compute_corr=True,
    )

    mu_scaled = mu * w[:, None]

    pm.MvStudentT(
        'obs',
        nu=6,
        mu=mu_scaled,
        chol=chol,
        observed=Y_scaled,
    )

    joint_trace = pm.sample(
        draws=1500,
        tune=1500,
        target_accept=0.95,
        random_seed=42,
        init='adapt_diag',
        nuts_sampler='numpyro',
    )

    return joint_model, joint_trace, standardization, Y, w

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In [ ]: # =====
# CELL 6: CONVERGENCE DIAGNOSTICS FUNCTIONS
# =====

def print_convergence_diagnostics(trace, label: str):
    print(f"\n{label} - Convergence Diagnostics")
    print("-" * 60)

    divergences = int(trace.sample_stats['diverging'].sum())
    print(f"    Divergences: {divergences} (OK if < 10)")
    if divergences > 10:
        print("    ⚠ WARNING: High divergence count")
    else:
        print("    ✓ Divergences acceptable")

    try:
        rhat = az.rhat(trace, var_names=['alpha', 'league_tax', 'beta_age'])
        rhat_values = []

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    for var_name in ['alpha', 'league_tax', 'beta_age']:
        if var_name in rhat:
            vals = rhat[var_name].values.flatten()
            rhat_values.extend(v for v in vals if np.isfinite(v))
    if rhat_values:
        rhat_max = float(np.max(rhat_values))
        print(f"    Max Rhat: {rhat_max:.4f}")
        if rhat_max > 1.05:
            print("    ⚠ WARNING: Some parameters have Rhat > 1.05")
        else:
            print("    ✓ All Rhat values acceptable")
except Exception as e:
    print(f"    Rhat computation failed: {e}")

print("-" * 60)

def plot_convergence_diagnostics(trace, label: str):
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))
    fig.suptitle(f'{label} - Convergence Diagnostics', fontsize=14, fontweight='bold')

    divergences = int(trace.sample_stats['diverging'].sum())
    ax = axes[0, 0]
    ax.bar(['Divergences'], [divergences],
           color='coral' if divergences > 10 else 'green', alpha=0.7)
    ax.axhline(y=10, color='red', linestyle='--', label='Warning threshold')
    ax.set_ylabel('Count')
    ax.set_title('Sampling Divergences')
    ax.legend()
    ax.grid(alpha=0.3, axis='y')

    ax = axes[0, 1]
    ess = az.ess(trace, var_names=['alpha']).to_array().values.flatten()
    ax.hist(ess, bins=20, color='steelblue', alpha=0.7, edgecolor='black')
    ax.set_xlabel('ESS')
    ax.set_ylabel('Frequency')
    ax.set_title('Effective Sample Size Distribution')
    ax.grid(alpha=0.3, axis='y')

    ax = axes[1, 0]
    try:
        rhat = az.rhat(trace, var_names=['alpha', 'league_tax', 'beta_age'])
        rhat_vals = []
        for var_name in ['alpha', 'league_tax', 'beta_age']:
            if var_name in rhat:
                vals = rhat[var_name].values.flatten()
                rhat_vals.extend(v for v in vals if np.isfinite(v))
        ax.hist(rhat_vals, bins=20, color='green', alpha=0.7, edgecolor='black')
        ax.axvline(x=1.05, color='red', linestyle='--', linewidth=2, label='Threshold (1.05)')
        ax.set_xlabel('Rhat')
        ax.set_ylabel('Frequency')
        ax.set_title('Rhat Distribution')
        ax.legend()
        ax.grid(alpha=0.3, axis='y')
    except Exception:
        ax.text(0.5, 0.5, 'Rhat computation unavailable', ha='center', va='center')

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ax = axes[1, 1]
alpha_traces = trace.posterior['alpha'].values
for i in range(min(4, alpha_traces.shape[-1])):
    ax.plot(alpha_traces[:, :, i].flatten(), alpha=0.7, linewidth=0.5)
ax.set_xlabel('Iteration')
ax.set_ylabel('Alpha values')
ax.set_title('Trace Plots (alpha parameters)')
ax.grid(alpha=0.3)

plt.tight_layout()
return fig

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In [ ]: # =====
# CELL 7: LOAD ATT+MID SEASON DATA
# =====
# Train seasons: 2017/18-2021/22 → 5 bridge pairs
# (2022/23→2023/24 and 2023/24→2024/25 are both held out as test cohorts)

train_seasons = [
    '2017_2018', '2018_2019', '2019_2020',
    '2020_2021', '2021_2022', '2022_2023', '2023_2024',
]

print("Loading season data for ATT+MID...")
data_att_mid = {}
for s in train_seasons:
    data_att_mid[s] = load_season_att_mid(
        f'players_data-{s}.csv',
        is_historical=True,
        stats_list=COMMON_STATS_ATT_MID,
    )

# Load held-out target season – modern format, never used in training bridge
df_2425_att_mid = load_season_att_mid(
    'players_data-2024_2025.csv',
    is_historical=False,
    stats_list=COMMON_STATS_ATT_MID,
)

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Loading season data for ATT+MID...

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In [ ]: # =====
# CELL 8: BUILD & FIT ATT+MID MODEL
# =====

print("\n" + "=" * 80)
print("BUILDING COMBINED ATT+MID BRIDGE")
print("=" * 80 + "\n")

valid_positions_att_mid = list(
    set(POSITION_FILTERS_ATT_MID['ATT'] + POSITION_FILTERS_ATT_MID['MID'])
)
print(f"Valid positions (ATT+MID): {valid_positions_att_mid}")

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df_bridge_att_mid,
age_mean_att_mid,
team_labels_att_mid,
pos_labels_att_mid,
log_ratio_cols_att_mid,
) = build_bridge(
    data_att_mid,
    valid_positions_att_mid,
    COMMON_STATS_ATT_MID,
    train_seasons,
    label='ATT+MID',
)

if df_bridge_att_mid is not None:
    n_teams_att_mid = len(team_labels_att_mid)
    n_pos_att_mid = len(pos_labels_att_mid)
    n_stats_att_mid = len(COMMON_STATS_ATT_MID)

    print(
        f" {len(df_bridge_att_mid)} transfers, "
        f"{n_teams_att_mid} teams, {n_pos_att_mid} position(s)"
    )
    print(f" Stats: {COMMON_STATS_ATT_MID}\n")

    print("=" * 80)
    print("FITTING JOINT MULTIVARIATE MODEL (ATT+MID)")
    print("=" * 80 + "\n")

    (
        joint_model_att_mid,
        joint_trace_att_mid,
        standardization_att_mid,
        Y_std_att_mid,
        w_att_mid,
    ) = fit_joint_model(
        df_bridge_att_mid,
        COMMON_STATS_ATT_MID,
        n_pos_att_mid,
        n_teams_att_mid,
    )
    print("✓ Model fitted (ATT+MID)\n")
    print_convergence_diagnostics(joint_trace_att_mid, "ATT+MID")

else:
    print("ERROR: Could not build bridge for ATT+MID")

# Save ATT+MID model artifacts
import pickle
import os

save_dir = '/content/model_artifacts'
os.makedirs(save_dir, exist_ok=True)

joint_trace_att_mid.to_netcdf(f'{save_dir}/trace_att_mid.nc')

with open(f'{save_dir}/artifacts_att_mid.pkl', 'wb') as f:

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pickle.dump({
    'standardization': standardization_att_mid,
    'age_mean':        age_mean_att_mid,
    'team_labels':     team_labels_att_mid,
    'pos_labels':      pos_labels_att_mid,
    'n_teams':         n_teams_att_mid,
    'n_pos':           n_pos_att_mid,
}, f)

print("✓ ATT+MID artifacts saved to", save_dir)

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BUILDING COMBINED ATT+MID BRIDGE
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Valid positions (ATT+MID): ['MID', 'FWD']
Skipping excluded pair: 2022_2023_to_2023_2024
Bridge (ATT+MID): 174 transfers after cleaning
Transfer seasons: ['2017_2018_to_2018_2019', '2018_2019_to_2019_2020', '2019_2020_to_2020_2021', '2020_2021_to_2021_2022', '2021_2022_to_2022_2023']
174 transfers, 26 teams, 2 position(s)
Stats: ['KP', 'PrgP', 'PrgC', 'SCA']

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FITTING JOINT MULTIVARIATE MODEL (ATT+MID)
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sample: 100%|██████████| 3000/3000 [03:13<00:00, 15.52it/s, 63 steps of size 9.75e-02. acc. prob=0.95]
sample: 100%|██████████| 3000/3000 [02:52<00:00, 17.35it/s, 31 steps of size 1.00e-01. acc. prob=0.93]
✓ Model fitted (ATT+MID)

```

ATT+MID - Convergence Diagnostics

```

-----
Divergences: 0 (OK if < 10)
✓ Divergences acceptable
Max Rhat: 1.0023
✓ All Rhat values acceptable
-----
✓ ATT+MID artifacts saved to /content/model_artifacts

```

```

In [ ]: # =====
# CELL 8b: SENSITIVITY ANALYSIS – DEGREES OF FREEDOM (nu) – ATT+MID
# =====

print("\n" + "=" * 80)
print("SENSITIVITY ANALYSIS: Degrees of Freedom (nu) – ATT+MID")
print("=" * 80 + "\n")

if df_bridge_att_mid is not None:
    nu_values      = [4, 6, 8, 10]
    nu_results_att = {}

    league_id_vals = df_bridge_att_mid['League_ID'].values.astype(int)

```

```

is_foreign_vals = df_bridge_att_mid['is_foreign'].values.astype(int)
pos_id_vals     = df_bridge_att_mid['pos_id'].values.astype(int)
team_id_vals    = df_bridge_att_mid['team_id'].values.astype(int)
age_c_vals      = (df_bridge_att_mid['Age_c'].values / 5.0).astype(float)
minutes_src_vals = df_bridge_att_mid['90s_src'].values
w_vals_main     = np.sqrt(minutes_src_vals / (minutes_src_vals + 5.0))
Y_scaled_att    = Y_std_att_mid * w_vals_main[:, None]

for nu_val in nu_values:
    print(f"Fitting ATT+MID model with nu={nu_val}...")
    with pm.Model() as nu_model:
        alpha_s = pm.Normal('alpha', 0, 0.2, shape=n_stats_att_mid)
        mu_lt_s = pm.Normal('mu_lt', 0, 0.2, shape=n_stats_att_mid)
        sigma_lt_s = pm.HalfStudentT('sigma_lt', nu=4, sigma=0.5, shape=n_stats_att_mid)
        z_league_s = pm.Normal('z_league', 0, 1, shape=(4, n_stats_att_mid))
        league_tax_s = pm.Deterministic(
            'league_tax', mu_lt_s[None, :] + z_league_s * sigma_lt_s[None, :]
        )
        sigma_pos_s = pm.HalfNormal('sigma_pos', 0.2)
        z_pos_s = pm.Normal('z_pos', 0, 1, shape=(n_pos_att_mid, n_stats_att_mid))
        pos_eff_s = pm.Deterministic('pos_eff', z_pos_s * sigma_pos_s)
        sigma_team_s = pm.HalfNormal('sigma_team', 0.2)
        z_team_s = pm.Normal('z_team', 0, 1, shape=(n_teams_att_mid, n_stats_att_mid))
        team_eff_s = pm.Deterministic('team_eff', z_team_s * sigma_team_s)
        beta_age_s = pm.Normal('beta_age', 0, 0.05, shape=n_stats_att_mid)
        beta_age2_s = pm.Normal('beta_age2', 0, 0.02, shape=n_stats_att_mid)
        safe_lid_s = pm.intX(np.clip(league_id_vals, 0, 3))
        mu_s = (
            alpha_s
            + is_foreign_vals[:, None] * league_tax_s[safe_lid_s]
            + pos_eff_s[pos_id_vals]
            + team_eff_s[team_id_vals]
            + beta_age_s * age_c_vals[:, None]
            + beta_age2_s * (age_c_vals[:, None] ** 2)
        )
        sd_dist_s = pm.HalfNormal.dist(0.3)
        chol_s, _, _ = pm.LKJCholeskyCov(
            'chol', n=n_stats_att_mid, eta=3.0,
            sd_dist=sd_dist_s, compute_corr=True,
        )
        pm.MvStudentT(
            'obs', nu=nu_val,
            mu=mu_s * w_vals_main[:, None],
            chol=chol_s,
            observed=Y_scaled_att,
        )
        nu_trace = pm.sample(
            draws=1000, tune=1000, target_accept=0.95,
            random_seed=42, init='adapt_diag',
            nuts_sampler='numpyro', progressbar=True,
        )

    lt_mean = nu_trace.posterior['league_tax'].mean(dim=('chain', 'draw')).values
    divergences = int(nu_trace.sample_stats['diverging'].sum())
    nu_results_att[nu_val] = {'league_tax_mean': lt_mean, 'divergences': divergences}
    print(f" nu={nu_val}: divergences={divergences}")

```

```

# index 0 = KP (first stat after removing npKG)
print("\nLeague tax sensitivity (ATT+MID, KP - 0=Ligue1,1=Bundesliga,2=SerieA,3=LaLiga)
print(f"{'nu':>5} | {'Ligue1':>8} {'Bundesliga':>10} {'Serie A':>8} {'La Liga':>8}")
print("-" * 50)
for nu_val in nu_values:
    lt = nu_results_att[nu_val]['league_tax_mean'][:, 0]
    print(f"{'nu_val':>5} | {lt[0]:>8.4f} {lt[1]:>10.4f} {lt[2]:>8.4f} {lt[3]:>8.4f}")
else:
    print("SKIPPED: ATT+MID bridge not available.")

```

=====

SENSITIVITY ANALYSIS: Degrees of Freedom (nu) – ATT+MID

=====

Fitting ATT+MID model with nu=4...

sample: 100%|██████████████████| 2000/2000 [02:30<00:00, 13.25it/s, 63 steps of size 7.66e-02. acc. prob=0.97]

sample: 100%|██████████████████| 2000/2000 [02:19<00:00, 14.33it/s, 63 steps of size 6.99e-02. acc. prob=0.96]

nu=4: divergences=0

Fitting ATT+MID model with nu=6...

sample: 100%|██████████████████| 2000/2000 [02:26<00:00, 13.63it/s, 63 steps of size 7.96e-02. acc. prob=0.96]

sample: 100%|██████████████████| 2000/2000 [02:19<00:00, 14.33it/s, 63 steps of size 8.40e-02. acc. prob=0.95]

nu=6: divergences=0

Fitting ATT+MID model with nu=8...

sample: 100%|██████████████████| 2000/2000 [02:23<00:00, 13.89it/s, 63 steps of size 9.43e-02. acc. prob=0.95]

sample: 100%|██████████████████| 2000/2000 [02:17<00:00, 14.51it/s, 63 steps of size 7.34e-02. acc. prob=0.97]

nu=8: divergences=0

Fitting ATT+MID model with nu=10...

sample: 100%|██████████████████| 2000/2000 [02:23<00:00, 13.97it/s, 63 steps of size 9.05e-02. acc. prob=0.96]

sample: 100%|██████████████████| 2000/2000 [02:18<00:00, 14.49it/s, 63 steps of size 8.52e-02. acc. prob=0.96]

nu=10: divergences=0

League tax sensitivity (ATT+MID, KP - 0=Ligue1,1=Bundesliga,2=SerieA,3=LaLiga):

nu	Ligue1	Bundesliga	Serie A	La Liga
4	-0.0141	-0.1848	-0.2011	0.1587
6	-0.0240	-0.1833	-0.2026	0.1409
8	-0.0304	-0.1792	-0.1993	0.1255
10	-0.0325	-0.1723	-0.1966	0.1098

```

In [ ]: # =====
# CELL 8c: SENSITIVITY ANALYSIS – WEIGHTING SCHEME – ATT+MID
# =====

print("\n" + "=" * 80)
print("SENSITIVITY ANALYSIS: Weighting Scheme – ATT+MID")
print("=" * 80 + "\n")

```

```

if df_bridge_att_mid is not None:
    weight_constants = [3, 5, 10]
    weight_results_att = {}

    for wc in weight_constants:
        print(f"Fitting ATT+MID model with weighting constant={wc}...")
        w_wc = np.sqrt(minutes_src_vals / (minutes_src_vals + wc))
        Y_scaled_wc = Y_std_att_mid * w_wc[:, None]

        with pm.Model() as w_model_att:
            alpha_w = pm.Normal('alpha', 0, 0.2, shape=n_stats_att_mid)
            mu_lt_w = pm.Normal('mu_lt', 0, 0.2, shape=n_stats_att_mid)
            sigma_lt_w = pm.HalfStudentT('sigma_lt', nu=4, sigma=0.5, shape=n_stats_att_mid)
            z_league_w = pm.Normal('z_league', 0, 1, shape=(4, n_stats_att_mid))
            league_tax_w = pm.Deterministic(
                'league_tax', mu_lt_w[None, :] + z_league_w * sigma_lt_w[None, :]
            )
            sigma_pos_w = pm.HalfNormal('sigma_pos', 0.2)
            z_pos_w = pm.Normal('z_pos', 0, 1, shape=(n_pos_att_mid, n_stats_att_mid))
            pos_eff_w = pm.Deterministic('pos_eff', z_pos_w * sigma_pos_w)
            sigma_team_w = pm.HalfNormal('sigma_team', 0.2)
            z_team_w = pm.Normal('z_team', 0, 1, shape=(n_teams_att_mid, n_stats_att_mid))
            team_eff_w = pm.Deterministic('team_eff', z_team_w * sigma_team_w)
            beta_age_w = pm.Normal('beta_age', 0, 0.05, shape=n_stats_att_mid)
            beta_age2_w = pm.Normal('beta_age2', 0, 0.02, shape=n_stats_att_mid)
            safe_lid_w = pm.intX(np.clip(league_id_vals, 0, 3))
            mu_w = (
                alpha_w
                + is_foreign_vals[:, None] * league_tax_w[safe_lid_w]
                + pos_eff_w[pos_id_vals]
                + team_eff_w[team_id_vals]
                + beta_age_w * age_c_vals[:, None]
                + beta_age2_w * (age_c_vals[:, None] ** 2)
            )
            sd_dist_w = pm.HalfNormal.dist(0.3)
            chol_w, _, _ = pm.LKJCholeskyCov(
                'chol', n=n_stats_att_mid, eta=3.0,
                sd_dist=sd_dist_w, compute_corr=True,
            )
            pm.MvStudentT(
                'obs', nu=6,
                mu=mu_w * w_wc[:, None],
                chol=chol_w,
                observed=Y_scaled_wc,
            )
            w_trace_att = pm.sample(
                draws=1000, tune=1000, target_accept=0.95,
                random_seed=42, init='adapt_diag',
                nuts_sampler='numpyro', progressbar=True,
            )

        lt_mean = w_trace_att.posterior['league_tax'].mean(dim=('chain', 'draw')).value
        divergences = int(w_trace_att.sample_stats['diverging'].sum())
        weight_results_att[wc] = {'league_tax_mean': lt_mean, 'divergences': divergences}
    print(f" wc={wc}: divergences={divergences}")

```

```

print("\nWeighting sensitivity (ATT+MID, KP - 0=Ligue1,1=Bundesliga,2=SerieA,3=LaLiga):
print(f"{'wc':>5} | {'Ligue1':>8} {'Bundesliga':>10} {'Serie A':>8} {'La Liga':>8}")
print("-" * 50)
for wc in weight_constants:
    lt = weight_results_att[wc]['league_tax_mean'][:, 0]
    print(f"{'wc':>5} | {lt[0]:>8.4f} {lt[1]:>10.4f} {lt[2]:>8.4f} {lt[3]:>8.4f}")
else:
    print("SKIPPED: ATT+MID bridge not available.")

```

===== SENSITIVITY ANALYSIS: Weighting Scheme - ATT+MID =====

Fitting ATT+MID model with weighting constant=3...

sample: 100%|██████████████████| 2000/2000 [02:23<00:00, 13.98it/s, 63 steps of size 8.13e-02. acc. prob=0.96]

sample: 100%|██████████████████| 2000/2000 [02:14<00:00, 14.86it/s, 63 steps of size 9.27e-02. acc. prob=0.95]

wc=3: divergences=0

Fitting ATT+MID model with weighting constant=5...

sample: 100%|██████████████████| 2000/2000 [02:26<00:00, 13.61it/s, 63 steps of size 7.05e-02. acc. prob=0.97]

sample: 100%|██████████████████| 2000/2000 [02:23<00:00, 13.91it/s, 63 steps of size 7.46e-02. acc. prob=0.97]

wc=5: divergences=0

Fitting ATT+MID model with weighting constant=10...

sample: 100%|██████████████████| 2000/2000 [02:26<00:00, 13.68it/s, 63 steps of size 9.59e-02. acc. prob=0.95]

sample: 100%|██████████████████| 2000/2000 [02:21<00:00, 14.09it/s, 63 steps of size 7.61e-02. acc. prob=0.96]

wc=10: divergences=0

Weighting sensitivity (ATT+MID, KP - 0=Ligue1,1=Bundesliga,2=SerieA,3=LaLiga):

wc	Ligue1	Bundesliga	Serie A	La Liga
3	-0.0201	-0.1849	-0.2021	0.1470
5	-0.0239	-0.1793	-0.1970	0.1362
10	-0.0258	-0.1850	-0.2066	0.1339

```

In [ ]: # =====
# CELL 8d: LEAVE-ONE-SEASON-OUT (LOSO) CROSS-VALIDATION - ATT+MID
# =====

print("\n" + "=" * 80)
print("LEAVE-ONE-SEASON-OUT (LOSO) CROSS-VALIDATION - ATT+MID")
print("=" * 80 + "\n")

loso_results_att = [] # fold-level MAE summary rows
serie_a_tax_folds = [] # Serie A SCA posterior stability monitor

if df_bridge_att_mid is not None:
    all_pairs_att = sorted(df_bridge_att_mid['Transfer_Season'].unique())
    print(f"Training bridge season pairs: {all_pairs_att}\n")
    sca_idx = COMMON_STATS_ATT_MID.index('SCA')

```



```

for held_out_pair in all_pairs_att:
    print(f"--- Fold: holding out {held_out_pair} ---")

    df_tr = df_bridge_att_mid[
        df_bridge_att_mid['Transfer_Season'] != held_out_pair
    ].copy().reset_index(drop=True)

    df_va = df_bridge_att_mid[
        df_bridge_att_mid['Transfer_Season'] == held_out_pair
    ].copy().reset_index(drop=True)

    if df_tr.empty or df_va.empty:
        print(" Skipping (empty fold)\n")
        continue

    # Re-factorize on this fold's training subset
    df_tr['team_id'], tl_f = pd.factorize(df_tr['Team'])
    df_tr['pos_id'], pl_f = pd.factorize(df_tr['Position'])
    n_teams_f = len(tl_f)
    n_pos_f = len(pl_f)
    n_stats_f = len(COMMON_STATS_ATT_MID)

    # Standardize on fold training data (never leak from validation)
    std_f = {}
    Y_f = np.zeros((len(df_tr), n_stats_f))
    for j, s in enumerate(COMMON_STATS_ATT_MID):
        vals = df_tr[f'{s}_log_ratio'].values
        m_, d_ = vals.mean(), vals.std()
        std_f[s] = {'mean': float(m_), 'std': float(d_)}
        Y_f[:, j] = (vals - m_) / (d_ + 1e-8)

    age_mean_f = df_tr['Age'].mean()
    df_tr['Age_c'] = df_tr['Age'] - age_mean_f

    mins_f = df_tr['90s_src'].values
    w_f = np.sqrt(mins_f / (mins_f + 5.0))
    Y_f_sc = Y_f * w_f[:, None]

    lid_f = df_tr['League_ID'].values.astype(int)
    isf_f = df_tr['is_foreign'].values.astype(int)
    pid_f = df_tr['pos_id'].values.astype(int)
    tid_f = df_tr['team_id'].values.astype(int)
    agec_f = (df_tr['Age_c'].values / 5.0).astype(float)

    with pm.Model() as loso_att_model:
        al = pm.Normal('alpha', 0, 0.2, shape=n_stats_f)
        ml = pm.Normal('mu_lt', 0, 0.2, shape=n_stats_f)
        sl = pm.HalfStudentT('sigma_lt', nu=4, sigma=0.5, shape=n_stats_f)
        zl = pm.Normal('z_league', 0, 1, shape=(4, n_stats_f))
        lt = pm.Deterministic('league_tax', ml[None, :] + zl * sl[None, :])

        if n_pos_f > 1:
            sp = pm.HalfNormal('sigma_pos', 0.2)
            zp = pm.Normal('z_pos', 0, 1, shape=(n_pos_f, n_stats_f))
            pe = pm.Deterministic('pos_eff', zp * sp)
        else:

```

```

        pe = 0

    st = pm.HalfNormal('sigma_team', 0.2)
    zt = pm.Normal('z_team', 0, 1, shape=(n_teams_f, n_stats_f))
    te = pm.Deterministic('team_eff', zt * st)

    ba = pm.Normal('beta_age', 0, 0.05, shape=n_stats_f)
    ba2 = pm.Normal('beta_age2', 0, 0.02, shape=n_stats_f)

    sli = pm.intX(np.clip(lid_f, 0, 3))
    pc = pe[pid_f] if n_pos_f > 1 else 0

    mu_l = (
        al
        + isf_f[:, None] * lt[sli]
        + pc
        + te[tid_f]
        + ba * agec_f[:, None]
        + ba2 * (agec_f[:, None] ** 2)
    )

    sd_l = pm.HalfNormal.dist(0.3)
    ch_l, _, _ = pm.LKJCholeskyCov(
        'chol', n=n_stats_f, eta=3.0,
        sd_dist=sd_l, compute_corr=True,
    )
    pm.MvStudentT('obs', nu=6, mu=mu_l * w_f[:, None], chol=ch_l, observed=Y_f_sc)

    loso_tr = pm.sample(
        draws=500, tune=500,
        target_accept=0.95, random_seed=42,
        init='adapt_diag', nuts_sampler='numpyro',
        progressbar=False,
    )

    # Point predictions on held-out fold
    am = loso_tr.posterior['alpha'].mean(dim=('chain', 'draw')).values
    lm = loso_tr.posterior['league_tax'].mean(dim=('chain', 'draw')).values
    bm = loso_tr.posterior['beta_age'].mean(dim=('chain', 'draw')).values
    b2m = loso_tr.posterior['beta_age2'].mean(dim=('chain', 'draw')).values

    fold_maes = {s: [] for s in COMMON_STATS_ATT_MID}
    for i_v in range(len(df_va)):
        age_v = (df_va.iloc[i_v]['Age'] - age_mean_f) / 5.0
        isf_v = int(df_va.iloc[i_v]['is_foreign'])
        l_v = int(np.clip(df_va.iloc[i_v]['League_ID'], 0, 3))
        tax_v = lm[l_v, :] if isf_v else np.zeros(n_stats_f)
        pred_v = am + tax_v + bm * age_v + b2m * (age_v ** 2)
        for j, s in enumerate(COMMON_STATS_ATT_MID):
            ms, ds = std_f[s]['mean'], std_f[s]['std']
            # de-standardize both true and predicted
            y_true = df_va.iloc[i_v][f'{s}_log_ratio']
            y_pred = pred_v[j] * ds + ms
            fold_maes[s].append(abs(y_true - y_pred))

    fold_row = {'Fold': held_out_pair, 'N_val': len(df_va)}

```

```

for s in COMMON_STATS_ATT_MID:
    fold_row[f'MAE_{s}'] = round(float(np.mean(fold_maes[s])), 4)
loso_results_att.append(fold_row)

# Serie A SCA league tax monitoring
lt_samp = loso_tr.posterior['league_tax'].values.reshape(-1, 4, n_stats_f)
sa_sca = lt_samp[:, 2, sca_idx] # league index 2 = Serie A
serie_a_tax_folds.append({
    'Fold': held_out_pair,
    'SerieA_SCA_mean': round(float(np.mean(sa_sca)), 4),
    'SerieA_SCA_sd': round(float(np.std(sa_sca, ddof=1)), 4),
    'SerieA_SCA_p5': round(float(np.percentile(sa_sca, 5)), 4),
    'SerieA_SCA_p95': round(float(np.percentile(sa_sca, 95)), 4),
})

divs = int(losotr.sample_stats['diverging'].sum())
print(f" N_train={len(df_tr)}, N_val={len(df_va)}, divergences={divs}")
print(f" Fold MAEs: { {s: fold_row[f'MAE_{s}'] for s in COMMON_STATS_ATT_MID} }\n")

if loso_results_att:
    df_loso_att = pd.DataFrame(loso_results_att)
    print("\nLOSO Cross-Validation Summary - ATT+MID:")
    print(df_loso_att.to_string(index=False))
    mae_cols = [f'MAE_{s}' for s in COMMON_STATS_ATT_MID]
    print("\nMean MAE across folds:")
    print(df_loso_att[mae_cols].mean().round(4).to_string())

    if serie_a_tax_folds:
        df_sa = pd.DataFrame(serie_a_tax_folds)
        print("\n\nSerie A SCA League Tax - Posterior Stability across LOSO Folds:")
        print(df_sa.to_string(index=False))
        print(
            f"\n Range of fold means: "
            f"{df_sa['SerieA_SCA_mean'].min():.4f} → "
            f"{df_sa['SerieA_SCA_mean'].max():.4f}"
        )
    else:
        print("SKIPPED: ATT+MID bridge not available.")

```

```
=====
LEAVE-ONE-SEASON-OUT (LOSO) CROSS-VALIDATION – ATT+MID
=====
```

Training bridge season pairs: ['2017_2018_to_2018_2019', '2018_2019_to_2019_2020', '2019_2020_to_2020_2021', '2020_2021_to_2021_2022', '2021_2022_to_2022_2023']

```
--- Fold: holding out 2017_2018_to_2018_2019 ---
    N_train=132, N_val=42, divergences=0
    Fold MAEs: {'KP': 0.3804, 'PrgP': 0.2515, 'PrgC': 0.3616, 'SCA': 0.2571}

--- Fold: holding out 2018_2019_to_2019_2020 ---
    N_train=148, N_val=26, divergences=0
    Fold MAEs: {'KP': 0.3161, 'PrgP': 0.2759, 'PrgC': 0.3896, 'SCA': 0.2853}

--- Fold: holding out 2019_2020_to_2020_2021 ---
    N_train=141, N_val=33, divergences=0
    Fold MAEs: {'KP': 0.4055, 'PrgP': 0.2481, 'PrgC': 0.3191, 'SCA': 0.3101}

--- Fold: holding out 2020_2021_to_2021_2022 ---
    N_train=143, N_val=31, divergences=0
    Fold MAEs: {'KP': 0.36, 'PrgP': 0.3394, 'PrgC': 0.2649, 'SCA': 0.2487}

--- Fold: holding out 2021_2022_to_2022_2023 ---
    N_train=132, N_val=42, divergences=0
    Fold MAEs: {'KP': 0.4555, 'PrgP': 0.3207, 'PrgC': 0.4556, 'SCA': 0.2942}
```

LOSO Cross-Validation Summary – ATT+MID:

	Fold	N_val	MAE_KP	MAE_PrgP	MAE_PrgC	MAE_SCA
2017_2018_to_2018_2019		42	0.3804	0.2515	0.3616	0.2571
2018_2019_to_2019_2020		26	0.3161	0.2759	0.3896	0.2853
2019_2020_to_2020_2021		33	0.4055	0.2481	0.3191	0.3101
2020_2021_to_2021_2022		31	0.3600	0.3394	0.2649	0.2487
2021_2022_to_2022_2023		42	0.4555	0.3207	0.4556	0.2942

Mean MAE across folds:

```
MAE_KP      0.3835
MAE_PrgP    0.2871
MAE_PrgC    0.3582
MAE_SCA     0.2791
```

Serie A SCA League Tax – Posterior Stability across LOSO Folds:

	Fold	SerieA_SCA_mean	SerieA_SCA_sd	SerieA_SCA_p5	SerieA_SCA_p95
2017_2018_to_2018_2019		-0.0409	0.1630	-0.3090	0.2178
2018_2019_to_2019_2020		-0.2108	0.1883	-0.5440	0.0960
2019_2020_to_2020_2021		-0.1553	0.1734	-0.4439	0.1230
2020_2021_to_2021_2022		-0.3041	0.2224	-0.6856	0.0511
2021_2022_to_2022_2023		-0.1095	0.2055	-0.4591	0.2167

Range of fold means: -0.3041 → -0.0409

```
In [ ]: # =====
# CELL 9: LOAD DEF SEASON DATA
# =====
```

```

print("\nLoading season data for DEF...")
data_def = {}
for s in train_seasons:
    data_def[s] = load_season_def(
        f'players_data-{s}.csv',
        is_historical=True,
        stats_list=COMMON_STATS_DEF,
    )

# Load held-out target season - modern format
df_2425_def = load_season_def(
    'players_data-2024_2025.csv',
    is_historical=False,
    stats_list=COMMON_STATS_DEF,
)

```

Loading season data for DEF...

```

In [ ]: # =====
# CELL 10: BUILD & FIT DEF MODEL
# =====

print("\n" + "=" * 80)
print("BUILDING DEF BRIDGE")
print("=" * 80 + "\n")

valid_positions_def = list(POSITION_FILTERS_DEF['DEF'])
print(f"Valid positions (DEF): {valid_positions_def}")

(
    df_bridge_def,
    age_mean_def,
    team_labels_def,
    pos_labels_def,
    log_ratio_cols_def,
) = build_bridge(
    data_def,
    valid_positions_def,
    COMMON_STATS_DEF,
    train_seasons,
    label='DEF',
)

if df_bridge_def is not None:
    n_teams_def = len(team_labels_def)
    n_pos_def = len(pos_labels_def)
    n_stats_def = len(COMMON_STATS_DEF)

    print(
        f" {len(df_bridge_def)} transfers, "
        f"{n_teams_def} teams, {n_pos_def} position(s)"
    )
    print(f" Stats: {COMMON_STATS_DEF}\n")

    print("=" * 80)

```

```

print("FITTING JOINT MULTIVARIATE MODEL (DEF)")
print("=" * 80 + "\n")

(
    joint_model_def,
    joint_trace_def,
    standardization_def,
    Y_std_def,
    w_def,
) = fit_joint_model(
    df_bridge_def,
    COMMON_STATS_DEF,
    n_pos_def,
    n_teams_def,
)
print("✓ Model fitted (DEF)\n")
print_convergence_diagnostics(joint_trace_def, "DEF")

else:
    print("ERROR: Could not build bridge for DEF")

# Save DEF model artifacts
if df_bridge_def is not None:
    joint_trace_def.to_netcdf(f'{save_dir}/trace_def.nc')

    with open(f'{save_dir}/artifacts_def.pkl', 'wb') as f:
        pickle.dump({
            'standardization': standardization_def,
            'age_mean': age_mean_def,
            'team_labels': team_labels_def,
            'pos_labels': pos_labels_def,
            'n_teams': n_teams_def,
            'n_pos': n_pos_def,
        }, f)

    print("✓ DEF artifacts saved to", save_dir)

```

```

=====
BUILDING DEF BRIDGE
=====

```

```

Valid positions (DEF): ['DEF']
Skipping excluded pair: 2022_2023_to_2023_2024
Bridge (DEF): 106 transfers after cleaning
Transfer seasons: ['2017_2018_to_2018_2019', '2018_2019_to_2019_2020', '2019_2020_to_2020_2021', '2020_2021_to_2021_2022', '2021_2022_to_2022_2023']
106 transfers, 23 teams, 1 position(s)
Stats: ['TklW', 'Int', 'Clr', 'Blocks']

```

```

=====
FITTING JOINT MULTIVARIATE MODEL (DEF)
=====

```

```
sample: 100%|██████████████████| 3000/3000 [01:52<00:00, 26.63it/s, 31 steps of size 1.23e-01. acc. prob=0.96]
sample: 100%|██████████████████| 3000/3000 [01:44<00:00, 28.61it/s, 31 steps of size 1.65e-01. acc. prob=0.91]
✓ Model fitted (DEF)
```

DEF - Convergence Diagnostics

```
-----
Divergences: 0 (OK if < 10)
✓ Divergences acceptable
Max Rhat: 1.0026
✓ All Rhat values acceptable
-----
```

✓ DEF artifacts saved to /content/model_artifacts

```
In [ ]: # Download artifacts
        from google.colab import files
        import zipfile

        zip_path = '/content/model_artifacts.zip'
        with zipfile.ZipFile(zip_path, 'w') as zf:
            for fname in os.listdir(save_dir):
                zf.write(os.path.join(save_dir, fname), fname)

        files.download(zip_path)
        print("✓ model_artifacts.zip downloaded")
```

✓ model_artifacts.zip downloaded

```
In [ ]: # =====
        # CELL 10b: SENSITIVITY ANALYSIS - DEGREES OF FREEDOM (nu) - DEF
        # =====

        print("\n" + "=" * 80)
        print("SENSITIVITY ANALYSIS: Degrees of Freedom (nu) - DEF")
        print("=" * 80 + "\n")

        if df_bridge_def is not None:
            nu_values = [4, 6, 8, 10]
            nu_results_def = {}

            league_id_vals_def = df_bridge_def['League_ID'].values.astype(int)
            is_foreign_vals_def = df_bridge_def['is_foreign'].values.astype(int)
            team_id_vals_def = df_bridge_def['team_id'].values.astype(int)
            age_c_vals_def = (df_bridge_def['Age_c'].values / 5.0).astype(float)
            minutes_src_vals_def = df_bridge_def['90s_src'].values
            w_vals_def_main = np.sqrt(minutes_src_vals_def / (minutes_src_vals_def + 5.0))
            Y_scaled_def = Y_std_def * w_vals_def_main[:, None]

            for nu_val in nu_values:
                print(f"Fitting DEF model with nu={nu_val}...")
                with pm.Model() as nu_model_def:
                    alpha_nd = pm.Normal('alpha', 0, 0.2, shape=n_stats_def)
                    mu_lt_nd = pm.Normal('mu_lt', 0, 0.2, shape=n_stats_def)
                    sigma_lt_nd = pm.HalfStudentT('sigma_lt', nu=4, sigma=0.5, shape=n_stats_def)
```

```

z_league_nd = pm.Normal('z_league', 0, 1, shape=(4, n_stats_def))
league_tax_nd = pm.Deterministic(
    'league_tax', mu_lt_nd[None, :] + z_league_nd * sigma_lt_nd[None, :]
)
sigma_team_nd = pm.HalfNormal('sigma_team', 0.2)
z_team_nd = pm.Normal('z_team', 0, 1, shape=(n_teams_def, n_stats_def))
team_eff_nd = pm.Deterministic('team_eff', z_team_nd * sigma_team_nd)
beta_age_nd = pm.Normal('beta_age', 0, 0.05, shape=n_stats_def)
beta_age2_nd = pm.Normal('beta_age2', 0, 0.02, shape=n_stats_def)
safe_lid_nd = pm.intX(np.clip(league_id_vals_def, 0, 3))
mu_nd = (
    alpha_nd
    + is_foreign_vals_def[:, None] * league_tax_nd[safe_lid_nd]
    + team_eff_nd[team_id_vals_def]
    + beta_age_nd * age_c_vals_def[:, None]
    + beta_age2_nd * (age_c_vals_def[:, None] ** 2)
)
sd_dist_nd = pm.HalfNormal.dist(0.3)
chol_nd, _, _ = pm.LKJCholeskyCov(
    'chol', n=n_stats_def, eta=3.0,
    sd_dist=sd_dist_nd, compute_corr=True,
)
pm.MvStudentT(
    'obs', nu=nu_val,
    mu=mu_nd * w_vals_def_main[:, None],
    chol=chol_nd,
    observed=Y_scaled_def,
)
nu_trace_def = pm.sample(
    draws=1000, tune=1000, target_accept=0.95,
    random_seed=42, init='adapt_diag',
    nuts_sampler='numpyro', progressbar=True,
)

lt_mean = nu_trace_def.posterior['league_tax'].mean(dim=('chain', 'draw')).values
divergences = int(nu_trace_def.sample_stats['diverging'].sum())
nu_results_def[nu_val] = {'league_tax_mean': lt_mean, 'divergences': divergences}
print(f"nu={nu_val}: divergences={divergences}")

print("\nLeague tax sensitivity (DEF, Tk1W - 0=Ligue1,1=Bundesliga,2=SerieA,3=LaLiga):"
print(f"{'nu':>5} | {'Ligue1':>8} {'Bundesliga':>10} {'Serie A':>8} {'La Liga':>8}")
print("-" * 50)
for nu_val in nu_values:
    lt = nu_results_def[nu_val]['league_tax_mean'][:, 0]
    print(f"{'nu_val':>5} | {lt[0]:>8.4f} {lt[1]:>10.4f} {lt[2]:>8.4f} {lt[3]:>8.4f}")
else:
    print("SKIPPED: DEF bridge not available.")

```

```

=====
SENSITIVITY ANALYSIS: Degrees of Freedom (nu) – DEF
=====

```

Fitting DEF model with nu=4...


```

sample: 100%|██████████████████| 2000/2000 [01:29<00:00, 22.45it/s, 31 steps of size 1.32e
-01. acc. prob=0.95]
sample: 100%|██████████████████| 2000/2000 [01:23<00:00, 24.01it/s, 31 steps of size 1.37e
-01. acc. prob=0.95]
nu=4: divergences=0
Fitting DEF model with nu=6...

sample: 100%|██████████████████| 2000/2000 [01:39<00:00, 20.15it/s, 31 steps of size 9.69e
-02. acc. prob=0.98]
sample: 100%|██████████████████| 2000/2000 [01:19<00:00, 25.23it/s, 31 steps of size 1.33e
-01. acc. prob=0.95]
nu=6: divergences=0
Fitting DEF model with nu=8...

sample: 100%|██████████████████| 2000/2000 [01:28<00:00, 22.64it/s, 31 steps of size 1.08e
-01. acc. prob=0.97]
sample: 100%|██████████████████| 2000/2000 [01:21<00:00, 24.59it/s, 31 steps of size 1.32e
-01. acc. prob=0.94]
nu=8: divergences=0
Fitting DEF model with nu=10...

sample: 100%|██████████████████| 2000/2000 [01:28<00:00, 22.63it/s, 31 steps of size 1.09e
-01. acc. prob=0.97]
sample: 100%|██████████████████| 2000/2000 [01:23<00:00, 24.05it/s, 31 steps of size 1.47e
-01. acc. prob=0.93]
nu=10: divergences=0

```

League tax sensitivity (DEF, TklW – 0=Ligue1,1=Bundesliga,2=SerieA,3=LaLiga):

nu	Ligue1	Bundesliga	Serie A	La Liga
4	0.3432	0.1768	0.4011	0.2401
6	0.3415	0.1853	0.3907	0.2555
8	0.3381	0.1951	0.3880	0.2673
10	0.3368	0.1939	0.3838	0.2719

```

In [ ]: # =====
# CELL 10c: SENSITIVITY ANALYSIS – WEIGHTING SCHEME – DEF
# =====

print("\n" + "=" * 80)
print("SENSITIVITY ANALYSIS: Weighting Scheme – DEF")
print("=" * 80 + "\n")

if df_bridge_def is not None:
    weight_constants = [3, 5, 10]
    weight_results_def = {}

    for wc in weight_constants:
        print(f"Fitting DEF model with weighting constant={wc}...")
        w_wc_def = np.sqrt(minutes_src_vals_def / (minutes_src_vals_def + wc))
        Y_scaled_def_wc = Y_std_def * w_wc_def[:, None]

        with pm.Model() as w_model_def:
            alpha_wd = pm.Normal('alpha', 0, 0.2, shape=n_stats_def)
            mu_lt_wd = pm.Normal('mu_lt', 0, 0.2, shape=n_stats_def)
            sigma_lt_wd = pm.HalfStudentT('sigma_lt', nu=4, sigma=0.5, shape=n_stats_def)
            z_league_wd = pm.Normal('z_league', 0, 1, shape=(4, n_stats_def))
            league_tax_wd = pm.Deterministic(

```

```

        'league_tax', mu_lt_wd[None, :] + z_league_wd * sigma_lt_wd[None, :]
    )
    sigma_team_wd = pm.HalfNormal('sigma_team', 0.2)
    z_team_wd = pm.Normal('z_team', 0, 1, shape=(n_teams_def, n_stats_def))
    team_eff_wd = pm.Deterministic('team_eff', z_team_wd * sigma_team_wd)
    beta_age_wd = pm.Normal('beta_age', 0, 0.05, shape=n_stats_def)
    beta_age2_wd = pm.Normal('beta_age2', 0, 0.02, shape=n_stats_def)
    safe_lid_wd = pm.intX(np.clip(league_id_vals_def, 0, 3))
    mu_wd = (
        alpha_wd
        + is_foreign_vals_def[:, None] * league_tax_wd[safe_lid_wd]
        + team_eff_wd[team_id_vals_def]
        + beta_age_wd * age_c_vals_def[:, None]
        + beta_age2_wd * (age_c_vals_def[:, None] ** 2)
    )
    sd_dist_wd = pm.HalfNormal.dist(0.3)
    chol_wd, _, _ = pm.LKJCholeskyCov(
        'chol', n=n_stats_def, eta=3.0,
        sd_dist=sd_dist_wd, compute_corr=True,
    )
    pm.MvStudentT(
        'obs', nu=6,
        mu=mu_wd * w_wc_def[:, None],
        chol=chol_wd,
        observed=Y_scaled_def_wc,
    )
    w_trace_def = pm.sample(
        draws=1000, tune=1000, target_accept=0.95,
        random_seed=42, init='adapt_diag',
        nuts_sampler='numpyro', progressbar=True,
    )

    lt_mean = w_trace_def.posterior['league_tax'].mean(dim=('chain', 'draw')).value
    divergences = int(w_trace_def.sample_stats['diverging'].sum())
    weight_results_def[wc] = {'league_tax_mean': lt_mean, 'divergences': divergences}
    print(f" wc={wc}: divergences={divergences}")

    print("\nWeighting sensitivity (DEF, TklW - 0=Ligue1,1=Bundesliga,2=SerieA,3=LaLiga):")
    print(f"{'wc':>5} | {'Ligue1':>8} {'Bundesliga':>10} {'Serie A':>8} {'La Liga':>8}")
    print("-" * 50)
    for wc in weight_constants:
        lt = weight_results_def[wc]['league_tax_mean'][:, 0]
        print(f"{'wc':>5} | {lt[0]:>8.4f} {lt[1]:>10.4f} {lt[2]:>8.4f} {lt[3]:>8.4f}")
    else:
        print("SKIPPED: DEF bridge not available.")

```

```

=====
SENSITIVITY ANALYSIS: Weighting Scheme - DEF
=====

```

Fitting DEF model with weighting constant=3...

```

sample: 100%|██████████████████| 2000/2000 [01:35<00:00, 20.98it/s, 31 steps of size 9.95e
-02. acc. prob=0.98]
sample: 100%|██████████████████| 2000/2000 [01:21<00:00, 24.57it/s, 31 steps of size 1.40e
-01. acc. prob=0.94]

```

```
wc=3: divergences=0
Fitting DEF model with weighting constant=5...
```

```
sample: 100%|██████████████████| 2000/2000 [01:26<00:00, 23.09it/s, 31 steps of size 1.16e-01. acc. prob=0.97]
sample: 100%|██████████████████| 2000/2000 [01:21<00:00, 24.63it/s, 31 steps of size 1.42e-01. acc. prob=0.94]
```

```
wc=5: divergences=0
Fitting DEF model with weighting constant=10...
```

```
sample: 100%|██████████████████| 2000/2000 [01:28<00:00, 22.66it/s, 31 steps of size 1.10e-01. acc. prob=0.97]
sample: 100%|██████████████████| 2000/2000 [01:42<00:00, 19.60it/s, 63 steps of size 7.73e-02. acc. prob=0.98]
```

```
wc=10: divergences=0
```

Weighting sensitivity (DEF, TklW - 0=Ligue1,1=Bundesliga,2=SerieA,3=Laliga):

wc	Ligue1	Bundesliga	Serie A	La Liga
3	0.3414	0.1878	0.3926	0.2586
5	0.3423	0.1865	0.3918	0.2547
10	0.3416	0.1850	0.3895	0.2533

```
In [ ]: # =====
# CELL 10d: LEAVE-ONE-SEASON-OUT (LOSO) CROSS-VALIDATION - DEF
# =====

print("\n" + "=" * 80)
print("LEAVE-ONE-SEASON-OUT (LOSO) CROSS-VALIDATION - DEF")
print("=" * 80 + "\n")

loso_results_def = []

if df_bridge_def is not None:
    all_pairs_def = sorted(df_bridge_def['Transfer_Season'].unique())
    print(f"Training bridge season pairs (DEF): {all_pairs_def}\n")

    for held_out_pair in all_pairs_def:
        print(f"--- Fold: holding out {held_out_pair} ---")

        df_tr_d = df_bridge_def[
            df_bridge_def['Transfer_Season'] != held_out_pair
        ].copy().reset_index(drop=True)

        df_va_d = df_bridge_def[
            df_bridge_def['Transfer_Season'] == held_out_pair
        ].copy().reset_index(drop=True)

        if df_tr_d.empty or df_va_d.empty:
            print(" Skipping (empty fold)\n")
            continue

        df_tr_d['team_id'], tl_fd = pd.factorize(df_tr_d['Team'])
        n_teams_fd = len(tl_fd)
        n_stats_fd = len(COMMON_STATS_DEF)

        std_fd = {}
```

```

Y_fd = np.zeros((len(df_tr_d), n_stats_fd))
for j, s in enumerate(COMMON_STATS_DEF):
    vals = df_tr_d[f'{s}_log_ratio'].values
    m_, d_ = vals.mean(), vals.std()
    std_fd[s] = {'mean': float(m_), 'std': float(d_)}
    Y_fd[:, j] = (vals - m_) / (d_ + 1e-8)

age_mean_fd = df_tr_d['Age'].mean()
df_tr_d['Age_c'] = df_tr_d['Age'] - age_mean_fd

mins_fd = df_tr_d['90s_src'].values
w_fd = np.sqrt(mins_fd / (mins_fd + 5.0))
Y_fd_sc = Y_fd * w_fd[:, None]

lid_fd = df_tr_d['League_ID'].values.astype(int)
isf_fd = df_tr_d['is_foreign'].values.astype(int)
tid_fd = df_tr_d['team_id'].values.astype(int)
agec_fd = (df_tr_d['Age_c'].values / 5.0).astype(float)

with pm.Model() as loso_def_model:
    al_d = pm.Normal('alpha', 0, 0.2, shape=n_stats_fd)
    ml_d = pm.Normal('mu_lt', 0, 0.2, shape=n_stats_fd)
    sl_d = pm.HalfStudentT('sigma_lt', nu=4, sigma=0.5, shape=n_stats_fd)
    zl_d = pm.Normal('z_league', 0, 1, shape=(4, n_stats_fd))
    lt_d = pm.Deterministic('league_tax', ml_d[None, :] + zl_d * sl_d[None, :])

    st_d = pm.HalfNormal('sigma_team', 0.2)
    zt_d = pm.Normal('z_team', 0, 1, shape=(n_teams_fd, n_stats_fd))
    te_d = pm.Deterministic('team_eff', zt_d * st_d)

    ba_d = pm.Normal('beta_age', 0, 0.05, shape=n_stats_fd)
    ba2_d = pm.Normal('beta_age2', 0, 0.02, shape=n_stats_fd)

    sli_d = pm.intX(np.clip(lid_fd, 0, 3))
    mu_ld = (
        al_d
        + isf_fd[:, None] * lt_d[sli_d]
        + te_d[tid_fd]
        + ba_d * agec_fd[:, None]
        + ba2_d * (agec_fd[:, None] ** 2)
    )

    sd_ld = pm.HalfNormal.dist(0.3)
    ch_ld, _, _ = pm.LKJCholeskyCov(
        'chol', n=n_stats_fd, eta=3.0,
        sd_dist=sd_ld, compute_corr=True,
    )
    pm.MvStudentT('obs', nu=6, mu=mu_ld * w_fd[:, None], chol=ch_ld, observed=Y_fd_

    loso_tr_d = pm.sample(
        draws=1000, tune=1000,
        target_accept=0.95, random_seed=42,
        init='adapt_diag', nuts_sampler='numpyro',
        progressbar=False,
    )

```

```

am_d = loso_tr_d.posterior['alpha'].mean(dim=('chain', 'draw')).values
lm_d = loso_tr_d.posterior['league_tax'].mean(dim=('chain', 'draw')).values
bm_d = loso_tr_d.posterior['beta_age'].mean(dim=('chain', 'draw')).values
b2m_d = loso_tr_d.posterior['beta_age2'].mean(dim=('chain', 'draw')).values

fold_maes_d = {s: [] for s in COMMON_STATS_DEF}
for i_v in range(len(df_va_d)):
    age_v = (df_va_d.iloc[i_v]['Age'] - age_mean_fd) / 5.0
    isf_v = int(df_va_d.iloc[i_v]['is_foreign'])
    l_v = int(np.clip(df_va_d.iloc[i_v]['League_ID'], 0, 3))
    tax_v = lm_d[l_v, :] if isf_v else np.zeros(n_stats_fd)
    pred_v = am_d + tax_v + bm_d * age_v + b2m_d * (age_v ** 2)
    for j, s in enumerate(COMMON_STATS_DEF):
        ms, ds = std_fd[s]['mean'], std_fd[s]['std']
        y_true = df_va_d.iloc[i_v][f'{s}_log_ratio']
        y_pred = pred_v[j] * ds + ms
        fold_maes_d[s].append(abs(y_true - y_pred))

fold_row_d = {'Fold': held_out_pair, 'N_val': len(df_va_d)}
for s in COMMON_STATS_DEF:
    fold_row_d[f'MAE_{s}'] = round(float(np.mean(fold_maes_d[s])), 4)
loso_results_def.append(fold_row_d)

divs_d = int(losotr_d.sample_stats['diverging'].sum())
print(f" N_train={len(df_tr_d)}, N_val={len(df_va_d)}, divergences={divs_d}")
print(f" Fold MAEs: { {s: fold_row_d[f'MAE_{s}'] for s in COMMON_STATS_DEF} }\n")

if loso_results_def:
    df_loso_def = pd.DataFrame(loso_results_def)
    print("\nLOSO Cross-Validation Summary - DEF:")
    print(df_loso_def.to_string(index=False))
    mae_cols_d = [f'MAE_{s}' for s in COMMON_STATS_DEF]
    print("\nMean MAE across folds:")
    print(df_loso_def[mae_cols_d].mean().round(4).to_string())
else:
    print("SKIPPED: DEF bridge not available.")

```

```
=====
LEAVE-ONE-SEASON-OUT (LOSO) CROSS-VALIDATION – DEF
=====
```

```
Training bridge season pairs (DEF): ['2017_2018_to_2018_2019', '2018_2019_to_2019_2020', '20
19_2020_to_2020_2021', '2020_2021_to_2021_2022', '2021_2022_to_2022_2023']
```

```
--- Fold: holding out 2017_2018_to_2018_2019 ---
```

```
  N_train=87, N_val=19, divergences=0
```

```
  Fold MAEs: {'TkIW': 0.4228, 'Int': 0.2012, 'Clr': 0.2349, 'Blocks': 0.3116}
```

```
--- Fold: holding out 2018_2019_to_2019_2020 ---
```

```
  N_train=94, N_val=12, divergences=0
```

```
  Fold MAEs: {'TkIW': 0.2334, 'Int': 0.3188, 'Clr': 0.3146, 'Blocks': 0.2841}
```

```
--- Fold: holding out 2019_2020_to_2020_2021 ---
```

```
  N_train=86, N_val=20, divergences=0
```

```
  Fold MAEs: {'TkIW': 0.4498, 'Int': 0.2786, 'Clr': 0.2723, 'Blocks': 0.399}
```

```
--- Fold: holding out 2020_2021_to_2021_2022 ---
```

```
  N_train=84, N_val=22, divergences=0
```

```
  Fold MAEs: {'TkIW': 0.3852, 'Int': 0.4438, 'Clr': 0.2595, 'Blocks': 0.23}
```

```
--- Fold: holding out 2021_2022_to_2022_2023 ---
```

```
  N_train=73, N_val=33, divergences=0
```

```
  Fold MAEs: {'TkIW': 0.3065, 'Int': 0.2556, 'Clr': 0.3128, 'Blocks': 0.273}
```

```
LOSO Cross-Validation Summary – DEF:
```

	Fold	N_val	MAE_TkIW	MAE_Int	MAE_Clr	MAE_Blocks
2017_2018_to_2018_2019		19	0.4228	0.2012	0.2349	0.3116
2018_2019_to_2019_2020		12	0.2334	0.3188	0.3146	0.2841
2019_2020_to_2020_2021		20	0.4498	0.2786	0.2723	0.3990
2020_2021_to_2021_2022		22	0.3852	0.4438	0.2595	0.2300
2021_2022_to_2022_2023		33	0.3065	0.2556	0.3128	0.2730

```
Mean MAE across folds:
```

```
MAE_TkIW      0.3595
```

```
MAE_Int       0.2996
```

```
MAE_Clr       0.2788
```

```
MAE_Blocks    0.2995
```

```
In [ ]: # =====
# CELL 11: BUILD HELD-OUT TEST BRIDGES (2022/23→2023/24 AND 2023/24→2024/25)
# =====
# Two held-out cohorts are combined into a single test set:
#   - Cohort 1: 2022/23 → 2023/24 (historical format, both seasons)
#   - Cohort 2: 2023/24 → 2024/25 (mixed format; 90s fix ensures unit consistency)
# Training bridge uses 2017/18→2021/22 (5 pairs), never including either test pair.

print("\n" + "=" * 80)
print("BUILDING HELD-OUT TEST BRIDGES")
print(" Cohort 1: 2022/23 → 2023/24")
print(" Cohort 2: 2023/24 → 2024/25")
print("=" * 80 + "\n")
```

```

EPS_TEST = 0.05

def build_test_bridge(
    df_src: pd.DataFrame,
    df_trg: pd.DataFrame,
    valid_positions: list,
    stats_list: list,
    label: str,
    cohort_label: str,
    known_team_labels=None,
) -> pd.DataFrame:
    df_s0 = (
        df_src
        .sort_values('90s', ascending=False)
        .drop_duplicates('Player_Key')
    )
    df_s1 = (
        df_trg
        .sort_values('90s', ascending=False)
        .drop_duplicates('Player_Key')
    )

    m = pd.merge(df_s0, df_s1, on='Player_Key', suffixes=('_src', '_trg'))

    print(f" [{cohort_label}] Merged rows: {len(m)}")
    if len(m) > 0:
        print(f"    Squad change:    {(m['Squad_src'] != m['Squad_trg']).sum()}")
        print(f"    To PL:                {(m['Comp_trg'] == 'Premier League').sum()}")
        print(f"    From non-PL:         {(m['Comp_src'] != 'Premier League').sum()}")
        print(f"    Src mins >= 5:       {(m['90s_src'] >= 5.0).sum()}")
        print(f"    Trg mins >= 5:       {(m['90s_trg'] >= 5.0).sum()}")
        print(f"    Valid position:      {(m['Pos_Bucket_src'].isin(valid_positions)).sum()}")

    valid = m[
        (m['Squad_src'] != m['Squad_trg']) &
        (m['Comp_src'] != 'Premier League') &
        (m['Comp_trg'] == 'Premier League') &
        (m['90s_src'] >= 5.0) &
        (m['90s_trg'] >= 5.0) &
        (m['Pos_Bucket_src'].isin(valid_positions))
    ].copy()

    print(f"    Valid transfers after all filters: {len(valid)}")

    if valid.empty:
        print(f"    WARNING: No valid test transfers found for {label} [{cohort_label}]")
        return pd.DataFrame()

    rows = []
    for _, r in valid.iterrows():
        src_league = r['Comp_src']
        row = {
            'Player':      r['Player_src'],
            'League_ID':    LEAGUE_MAP.get(src_league, -1),
            'League_origin': src_league,
            'is_foreign':    int(src_league != 'Premier League'),

```

```

        'Age':          r['Age_src'],
        'Position':    r['Pos_Bucket_src'],
        'Team':         r['Squad_trg'],
        '90s_src':      r['90s_src'],
        'Cohort':       cohort_label,
    }
    for s in stats_list:
        src_val = r.get(f'{s}_src', np.nan)
        trg_val = r.get(f'{s}_trg', np.nan)
        src_rate = src_val / r['90s_src']
        trg_rate = trg_val / r['90s_trg']
        row[f'{s}_src'] = src_rate
        row[f'{s}_trg'] = trg_rate
        row[f'{s}_log_ratio'] = np.log((trg_rate + EPS_TEST) / (src_rate + EPS_TEST))
    rows.append(row)

df_test = pd.DataFrame(rows).dropna(subset=['Age'])

log_ratio_cols = [f'{s}_log_ratio' for s in stats_list]
for col in log_ratio_cols:
    if col in df_test.columns:
        df_test[col] = df_test[col].replace([np.inf, -np.inf], np.nan)
df_test = df_test.dropna(subset=log_ratio_cols)

if df_test.empty:
    print(f"    WARNING: No valid test transfers after cleaning for {label} [{cohort_label}]")
    return pd.DataFrame()

if known_team_labels is not None:
    known_teams = set(known_team_labels)
    unseen_clubs = sorted(df_test[~df_test['Team'].isin(known_teams)]['Team'].unique())
    if unseen_clubs:
        print(f"    NOTE: {len(unseen_clubs)} unseen club(s) - team_eff zeroed: {unseen_clubs}")
    else:
        print(f"    All destination clubs seen in training.")

print(f"    Test bridge ({label}, {cohort_label}): {len(df_test)} transfers\n")
return df_test.reset_index(drop=True)

# --- Cohort 1: 2022/23 → 2023/24 (both historical) ---
print("--- Cohort 1: 2022/23 → 2023/24 ---")
df_test_att_c1 = build_test_bridge(
    df_src=data_att_mid['2022_2023'],
    df_trg=data_att_mid['2023_2024'],
    valid_positions=valid_positions_att_mid,
    stats_list=COMMON_STATS_ATT_MID,
    label='ATT+MID',
    cohort_label='2022/23→2023/24',
    known_team_labels=team_labels_att_mid,
)

df_test_def_c1 = build_test_bridge(
    df_src=data_def['2022_2023'],
    df_trg=data_def['2023_2024'],
    valid_positions=valid_positions_def,

```



```

stats_list=COMMON_STATS_DEF,
label='DEF',
cohort_label='2022/23→2023/24',
known_team_labels=team_labels_def,
)

# --- Cohort 2: 2023/24 → 2024/25 (mixed format, 90s fix applied) ---
print("--- Cohort 2: 2023/24 → 2024/25 ---")
df_test_att_c2 = build_test_bridge(
    df_src=data_att_mid['2023_2024'],
    df_trg=df_2425_att_mid,
    valid_positions=valid_positions_att_mid,
    stats_list=COMMON_STATS_ATT_MID,
    label='ATT+MID',
    cohort_label='2023/24→2024/25',
    known_team_labels=team_labels_att_mid,
)

df_test_def_c2 = build_test_bridge(
    df_src=data_def['2023_2024'],
    df_trg=df_2425_def,
    valid_positions=valid_positions_def,
    stats_list=COMMON_STATS_DEF,
    label='DEF',
    cohort_label='2023/24→2024/25',
    known_team_labels=team_labels_def,
)

# --- Combine cohorts ---
df_test_att = pd.concat(
    [df for df in [df_test_att_c1, df_test_att_c2] if not df.empty],
    ignore_index=True,
)

df_test_def = pd.concat(
    [df for df in [df_test_def_c1, df_test_def_c2] if not df.empty],
    ignore_index=True,
)

print("=" * 60)
print(f"Combined ATT+MID test set: n={len(df_test_att)}")
print(f" Cohort 1 (2022/23→2023/24): {len(df_test_att_c1)}")
print(f" Cohort 2 (2023/24→2024/25): {len(df_test_att_c2)}")
print(f"\nCombined DEF test set:      n={len(df_test_def)}")
print(f" Cohort 1 (2022/23→2023/24): {len(df_test_def_c1)}")
print(f" Cohort 2 (2023/24→2024/25): {len(df_test_def_c2)}")
print("=" * 60)

```

=====

BUILDING HELD-OUT TEST BRIDGES

Cohort 1: 2022/23 → 2023/24

Cohort 2: 2023/24 → 2024/25

=====

--- Cohort 1: 2022/23 → 2023/24 ---

[2022/23→2023/24] Merged rows: 1502

Squad change: 416

To PL: 330

From non-PL: 1183

Src mins >= 5: 1439

Trg mins >= 5: 1400

Valid position: 802

Valid transfers after all filters: 27

NOTE: 2 unseen club(s) – team_eff zeroed: ['Burnley', 'Luton Town']

Test bridge (ATT+MID, 2022/23→2023/24): 27 transfers

[2022/23→2023/24] Merged rows: 587

Squad change: 135

To PL: 126

From non-PL: 461

Src mins >= 5: 568

Trg mins >= 5: 558

Valid position: 587

Valid transfers after all filters: 17

NOTE: 2 unseen club(s) – team_eff zeroed: ['Luton Town', 'Sheffield Utd']

Test bridge (DEF, 2022/23→2023/24): 17 transfers

--- Cohort 2: 2023/24 → 2024/25 ---

[2023/24→2024/25] Merged rows: 1486

Squad change: 399

To PL: 328

From non-PL: 1166

Src mins >= 5: 1403

Trg mins >= 5: 1390

Valid position: 800

Valid transfers after all filters: 18

NOTE: 1 unseen club(s) – team_eff zeroed: ['Ipswich Town']

Test bridge (ATT+MID, 2023/24→2024/25): 18 transfers

[2023/24→2024/25] Merged rows: 562

Squad change: 143

To PL: 123

From non-PL: 445

Src mins >= 5: 543

Trg mins >= 5: 532

Valid position: 562

Valid transfers after all filters: 18

All destination clubs seen in training.

Test bridge (DEF, 2023/24→2024/25): 18 transfers

=====

Combined ATT+MID test set: n=45

Cohort 1 (2022/23→2023/24): 27

Cohort 2 (2023/24→2024/25): 18

Combined DEF test set: n=35
Cohort 1 (2022/23→2023/24): 17
Cohort 2 (2023/24→2024/25): 18

=====

```
In [ ]: # =====
# CELL 12: BAYESIAN OUT-OF-SAMPLE VALIDATION
# =====

print("\n" + "=" * 80)
print("BAYESIAN OUT-OF-SAMPLE VALIDATION")
print("=" * 80 + "\n")

def standardize_test_bridge(df_test, stats_list, standardization):
    """Apply training-set standardization parameters to the test bridge."""
    n_stats = len(stats_list)
    Y_test = np.zeros((len(df_test), n_stats))
    for j, s in enumerate(stats_list):
        mean = standardization[s]['mean']
        std = standardization[s]['std']
        Y_test[:, j] = (df_test[f'{s}_log_ratio'].values - mean) / (std + 1e-8)
    return Y_test

def predict_bayesian_test(df_test, stats_list, standardization, joint_trace, age_mean):
    """
    Full posterior predictive intervals: for each observation, sample mu from
    posterior parameters then add MvStudentT observation noise via the posterior
    Cholesky. Propagates both parameter uncertainty and observation noise.
    """
    n_obs = len(df_test)
    n_stats = len(stats_list)

    alpha_samples = joint_trace.posterior['alpha'].values.reshape(-1, n_stats)
    league_tax_samp = joint_trace.posterior['league_tax'].values.reshape(-1, 4, n_stats)
    beta_age_samp = joint_trace.posterior['beta_age'].values.reshape(-1, n_stats)
    beta_age2_samp = joint_trace.posterior['beta_age2'].values.reshape(-1, n_stats)

    chol_raw = joint_trace.posterior['chol'].values
    n_chol = n_stats * (n_stats + 1) // 2
    chol_flat = chol_raw.reshape(-1, n_chol)
    n_samples = chol_flat.shape[0]

    chol_samp = np.zeros((n_samples, n_stats, n_stats))
    tril_idx = np.tril_indices(n_stats)
    chol_samp[:, tril_idx[0], tril_idx[1]] = chol_flat

    pred_mean = np.zeros((n_obs, n_stats))
    pred_p5 = np.zeros((n_obs, n_stats))
    pred_p95 = np.zeros((n_obs, n_stats))

    rng = np.random.default_rng(42)
    nu = 6
```

```

for i in range(n_obs):
    age_c = (df_test.iloc[i]['Age'] - age_mean) / 5.0
    is_for = int(df_test.iloc[i]['is_foreign'])
    l_idx = int(np.clip(df_test.iloc[i]['League_ID'], 0, 3))

    age_eff = beta_age_samp * age_c + beta_age2_samp * (age_c ** 2)
    tax = league_tax_samp[:, l_idx, :] if is_for else np.zeros((n_samples, n_stats))
    mu_samp = alpha_samples + tax + age_eff

    chi2_i = rng.chisquare(nu, size=n_samples) / nu
    z_i = rng.standard_normal(size=(n_samples, n_stats))
    t_i = z_i / np.sqrt(chi2_i[:, None])
    noise_i = np.einsum('sij,sj->si', chol_samp, t_i)

    y_pred_samples = mu_samp + noise_i
    pred_mean[i] = y_pred_samples.mean(axis=0)
    hdi_i = np.array([az.hdi(y_pred_samples[:, j], hdi_prob=0.90)
                      for j in range(n_stats)])
    pred_p5[i] = hdi_i[:, 0]
    pred_p95[i] = hdi_i[:, 1]

return pred_mean, pred_p5, pred_p95

def compute_validation_metrics(df_test, Y_test_std,
                              pred_mean_std, pred_p5_std, pred_p95_std,
                              stats_list, standardization, model_label):
    results = []
    for j, stat in enumerate(stats_list):
        mean_s = standardization[stat]['mean']
        std_s = standardization[stat]['std']

        y_true = Y_test_std[:, j] * std_s + mean_s
        y_pred = pred_mean_std[:, j] * std_s + mean_s
        y_lo = pred_p5_std[:, j] * std_s + mean_s
        y_hi = pred_p95_std[:, j] * std_s + mean_s

        results.append({
            'Stat': stat,
            'Model': model_label,
            'MAE': round(float(np.mean(np.abs(y_true - y_pred))), 4),
            'RMSE': round(float(np.sqrt(np.mean((y_true - y_pred) ** 2))), 4),
            'Coverage_90': round(float(np.mean((y_true >= y_lo) & (y_true <= y_hi))), 3),
            'N_test': len(df_test),
        })
    return pd.DataFrame(results)

def compute_univariate_baseline(df_test, Y_test_std, stats_list,
                               standardization, joint_trace, age_mean, df_train):
    """
    Univariate baseline: posterior-mean point prediction with prediction intervals
    based on out-of-sample residual std computed on the training set.
    """
    n_obs = len(df_test)
    n_stats = len(stats_list)

```

```

n_train = len(df_train)

alpha_mean = joint_trace.posterior['alpha'].mean(dim=('chain', 'draw')).values
lt_mean    = joint_trace.posterior['league_tax'].mean(dim=('chain', 'draw')).values
ba_mean    = joint_trace.posterior['beta_age'].mean(dim=('chain', 'draw')).values
ba2_mean   = joint_trace.posterior['beta_age2'].mean(dim=('chain', 'draw')).values

train_pred = np.zeros((n_train, n_stats))
for i in range(n_train):
    age_c    = (df_train.iloc[i]['Age'] - age_mean) / 5.0
    is_for   = int(df_train.iloc[i]['is_foreign'])
    l_idx    = int(np.clip(df_train.iloc[i]['League_ID'], 0, 3))
    tax      = lt_mean[l_idx, :] if is_for else np.zeros(n_stats)
    age_eff  = ba_mean * age_c + ba2_mean * (age_c ** 2)
    train_pred[i] = alpha_mean + tax + age_eff

Y_train_std = np.zeros((n_train, n_stats))
for j, s in enumerate(stats_list):
    mean_s = standardization[s]['mean']
    std_s  = standardization[s]['std']
    Y_train_std[:, j] = (df_train[f'{s}_log_ratio'].values - mean_s) / (std_s + 1e-8)

train_resid_std = (Y_train_std - train_pred).std(axis=0, ddof=1)

pred_mean_test = np.zeros((n_obs, n_stats))
for i in range(n_obs):
    age_c    = (df_test.iloc[i]['Age'] - age_mean) / 5.0
    is_for   = int(df_test.iloc[i]['is_foreign'])
    l_idx    = int(np.clip(df_test.iloc[i]['League_ID'], 0, 3))
    tax      = lt_mean[l_idx, :] if is_for else np.zeros(n_stats)
    age_eff  = ba_mean * age_c + ba2_mean * (age_c ** 2)
    pred_mean_test[i] = alpha_mean + tax + age_eff

pred_p5 = pred_mean_test - 1.64 * train_resid_std
pred_p95 = pred_mean_test + 1.64 * train_resid_std

return compute_validation_metrics(
    df_test, Y_test_std,
    pred_mean_test, pred_p5, pred_p95,
    stats_list, standardization, 'Baseline: Univariate',
)

# --- ATT+MID ---
if not df_test_att.empty:
    Y_test_att = standardize_test_bridge(
        df_test_att, COMMON_STATS_ATT_MID, standardization_att_mid
    )
    pred_mean_att, pred_p5_att, pred_p95_att = predict_bayesian_test(
        df_test_att, COMMON_STATS_ATT_MID, standardization_att_mid,
        joint_trace_att_mid, age_mean_att_mid,
    )
    val_hb_att = compute_validation_metrics(
        df_test_att, Y_test_att,
        pred_mean_att, pred_p5_att, pred_p95_att,
        COMMON_STATS_ATT_MID, standardization_att_mid, 'HBFLT (Hierarchical)',

```

```

    )
    val_uni_att = compute_univariate_baseline(
        df_test_att, Y_test_att,
        COMMON_STATS_ATT_MID, standardization_att_mid,
        joint_trace_att_mid, age_mean_att_mid,
        df_bridge_att_mid,
    )
    print("Bayesian Validation – ATT+MID (held-out 2023/2024 & 2024/25):")
    print(pd.concat([val_hb_att, val_uni_att]).sort_values('Stat').to_string(index=False))
    print()

# --- DEF ---
if not df_test_def.empty:
    Y_test_def = standardize_test_bridge(
        df_test_def, COMMON_STATS_DEF, standardization_def
    )
    pred_mean_def, pred_p5_def, pred_p95_def = predict_bayesian_test(
        df_test_def, COMMON_STATS_DEF, standardization_def,
        joint_trace_def, age_mean_def,
    )
    val_hb_def = compute_validation_metrics(
        df_test_def, Y_test_def,
        pred_mean_def, pred_p5_def, pred_p95_def,
        COMMON_STATS_DEF, standardization_def, 'HBFLT (Hierarchical)',
    )
    val_uni_def = compute_univariate_baseline(
        df_test_def, Y_test_def,
        COMMON_STATS_DEF, standardization_def,
        joint_trace_def, age_mean_def,
        df_bridge_def,
    )
    print("Bayesian Validation – DEF (held-out 2023/2024 & 2024/25):")
    print(pd.concat([val_hb_def, val_uni_def]).sort_values('Stat').to_string(index=False))
    print()

```

=====

BAYESIAN OUT-OF-SAMPLE VALIDATION

=====

Bayesian Validation – ATT+MID (held-out 2023/2024 & 2024/25):

Stat	Model	MAE	RMSE	Coverage_90	N_test
KP HBFLT (Hierarchical)		0.3447	0.4621	0.933	45
KP Baseline: Univariate		0.3444	0.4629	0.956	45
PrgC HBFLT (Hierarchical)		0.3387	0.4218	0.889	45
PrgC Baseline: Univariate		0.3370	0.4199	0.956	45
PrgP HBFLT (Hierarchical)		0.2992	0.4283	0.889	45
PrgP Baseline: Univariate		0.2983	0.4280	0.889	45
SCA HBFLT (Hierarchical)		0.2516	0.3293	0.911	45
SCA Baseline: Univariate		0.2505	0.3278	0.956	45

Bayesian Validation – DEF (held-out 2023/2024 & 2024/25):

Stat	Model	MAE	RMSE	Coverage_90	N_test
Blocks HBFLT (Hierarchical)		0.2843	0.3664	0.857	35
Blocks Baseline: Univariate		0.2833	0.3658	0.914	35
Clr HBFLT (Hierarchical)		0.3338	0.4415	0.800	35
Clr Baseline: Univariate		0.3318	0.4404	0.886	35
Int HBFLT (Hierarchical)		0.3869	0.4894	0.629	35
Int Baseline: Univariate		0.3855	0.4886	0.829	35
TklW HBFLT (Hierarchical)		0.2474	0.3074	0.943	35
TklW Baseline: Univariate		0.2453	0.3064	0.943	35

```
In [ ]: # =====
# CELL 13: EDA – INTERCEPTIONS (Int) UNDER-COVERAGE DIAGNOSTIC (DEF)
# =====
# Investigates under-coverage for Int (0.686).
# Hypothesis: Int log-ratios have higher variance in the test set than training
# (test std=0.486 vs training std=0.380), causing intervals to be too narrow
# for the left tail of the test distribution.

print("\n" + "=" * 80)
print("EDA: INTERCEPTIONS (Int) UNDER-COVERAGE DIAGNOSTIC (DEF)")
print("=" * 80 + "\n")

import scipy.stats as stats

TARGET_STATS_DEF = ['Int']

# ---- 1. Log-ratio distribution: training vs test ----
print("--- Log-ratio distribution: training bridge vs test bridge ---")
print(f"  {'Stat':<8} {'Train mean':>12} {'Train std':>12} {'Test mean':>12} {'Test std':>12}")
print("  " + "-"*60)
for s in COMMON_STATS_DEF:
    tr = df_bridge_def[f'{s}_log_ratio'].values
    te = df_test_def[f'{s}_log_ratio'].values
    print(f"  {s:<8} {tr.mean():>12.4f} {tr.std():>12.4f} {te.mean():>12.4f} {te.std():>12.4f}")

print()

# ---- 2. Full distributional stats for Int ----
for s in TARGET_STATS_DEF:
```

```

print(f"--- {s} detailed distribution ---")
for bridge_df, bridge_label in [
    (df_bridge_def, 'Training Bridge'),
    (df_test_def, 'Test Bridge'),
]:
    vals = bridge_df[f'{s}_log_ratio'].values
    print(f" {bridge_label} (n={len(vals)}):")
    print(f"     Mean={vals.mean():.4f} Std={vals.std():.4f} "
          f"Median={np.median(vals):.4f} "
          f"IQR=[{np.percentile(vals,25):.3f}, {np.percentile(vals,75):.3f}] "
          f"Min/Max=[{vals.min():.3f}, {vals.max():.3f}]")
    print(f"     Skew={stats.skew(vals):.4f} Kurt={stats.kurtosis(vals):.4f}")
print()

# ---- 3. Per-league log-ratio means (training bridge) ----
for s in TARGET_STATS_DEF:
    print(f"--- Per-league {s} log-ratio means (training bridge) ---")
    for league, grp in df_bridge_def.groupby('League_origin'):
        v = grp[f'{s}_log_ratio'].values
        print(f" {league:<15} n={len(v):3d} mean={v.mean():.4f} std={v.std():.4f}")
    print()

# ---- 4. Prediction interval analysis ----
print("--- Prediction interval widths vs true spread ---")
print(f" {'Stat':<8} {'Mean width':>12} {'True std':>12} {'Width/2 / std':>15} {'Coverage'
print(" " + "-"*60)
for s in TARGET_STATS_DEF:
    s_idx = COMMON_STATS_DEF.index(s)
    mean_s = standardization_def[s]['mean']
    std_s = standardization_def[s]['std']
    lo = pred_p5_def[:, s_idx] * std_s + mean_s
    hi = pred_p95_def[:, s_idx] * std_s + mean_s
    true = Y_test_def[:, s_idx] * std_s + mean_s
    widths = hi - lo
    cov = float(np.mean((true >= lo) & (true <= hi)))
    print(f" {s:<8} {widths.mean():>12.4f} {true.std():>12.4f} "
          f"{(widths.mean()/2)/true.std():>15.2f}x {cov:>10.3f}")

print()

# ---- 5. Per-observation detail for Int ----
for s in TARGET_STATS_DEF:
    s_idx = COMMON_STATS_DEF.index(s)
    mean_s = standardization_def[s]['mean']
    std_s = standardization_def[s]['std']
    lo = pred_p5_def[:, s_idx] * std_s + mean_s
    hi = pred_p95_def[:, s_idx] * std_s + mean_s
    true = Y_test_def[:, s_idx] * std_s + mean_s
    pred = pred_mean_def[:, s_idx] * std_s + mean_s

    print(f"--- {s} per-observation detail ---")
    print(f" {'Player':<25} {'Cohort':<20} {'True':>7} {'Pred':>7} "
          f"{ 'Lo':>7} {'Hi':>7} {'In?':>5}")
    for i in range(len(df_test_def)):
        player = df_test_def.iloc[i]['Player']
        cohort = df_test_def.iloc[i]['Cohort']

```



```

        covered = '✓' if lo[i] <= true[i] <= hi[i] else '✗'
        print(f" {player:<25} {cohort:<20} {true[i]:>7.4f} {pred[i]:>7.4f} "
              f"{lo[i]:>7.4f} {hi[i]:>7.4f} {covered:>5}")
    print()

# ---- 6. Coverage breakdown by cohort ----
print("---- Coverage breakdown by cohort (Int) ----")
for s in TARGET_STATS_DEF:
    s_idx = COMMON_STATS_DEF.index(s)
    mean_s = standardization_def[s]['mean']
    std_s = standardization_def[s]['std']
    lo = pred_p5_def[:, s_idx] * std_s + mean_s
    hi = pred_p95_def[:, s_idx] * std_s + mean_s
    true = Y_test_def[:, s_idx] * std_s + mean_s
    covered = (true >= lo) & (true <= hi)

    print(f" {s}:")
    for cohort in df_test_def['Cohort'].unique():
        mask = df_test_def['Cohort'].values == cohort
        cov = covered[mask].mean()
        n = mask.sum()
        print(f" {cohort}: {cov:.3f} ({covered[mask].sum()}/{n})")
    print()

# ---- 7. Plot: log-ratio distributions training vs test ----
fig, axes = plt.subplots(2, 4, figsize=(18, 8))
fig.suptitle('DEF Log-Ratio Distributions: Training vs Test', fontsize=13, fontweight='bold')

for col_idx, s in enumerate(COMMON_STATS_DEF):
    tr_vals = df_bridge_def[f'{s}_log_ratio'].values
    te_vals = df_test_def[f'{s}_log_ratio'].values

    # Training
    ax = axes[0, col_idx]
    highlight = s in TARGET_STATS_DEF
    ax.hist(tr_vals, bins=25, color='coral' if highlight else 'steelblue',
            alpha=0.7, edgecolor='black')
    ax.axvline(0, color='black', linestyle='--', linewidth=1.5)
    ax.axvline(tr_vals.mean(), color='red', linestyle='-', linewidth=1.5,
               label=f'mean={tr_vals.mean():.3f}')
    ax.set_title(f'{s} - Training (std={tr_vals.std():.3f})', fontsize=10)
    ax.set_xlabel('Log-ratio')
    ax.set_ylabel('Count')
    ax.legend(fontsize=8)
    ax.grid(alpha=0.3, axis='y')

    # Test
    ax = axes[1, col_idx]
    ax.hist(te_vals, bins=15, color='coral' if highlight else 'steelblue',
            alpha=0.7, edgecolor='black')
    ax.axvline(0, color='black', linestyle='--', linewidth=1.5)
    ax.axvline(te_vals.mean(), color='red', linestyle='-', linewidth=1.5,
               label=f'mean={te_vals.mean():.3f}')
    ax.set_title(f'{s} - Test (std={te_vals.std():.3f})', fontsize=10)
    ax.set_xlabel('Log-ratio')
    ax.set_ylabel('Count')

```

```
ax.legend(fontsize=8)
ax.grid(alpha=0.3, axis='y')

axes[0, 0].set_ylabel('Training\nCount', fontsize=10)
axes[1, 0].set_ylabel('Test\nCount', fontsize=10)

plt.tight_layout()
plt.savefig('appendix_def_log_ratio_distributions.pdf', bbox_inches='tight', dpi=300)
plt.savefig('appendix_def_log_ratio_distributions.png', bbox_inches='tight', dpi=300)
plt.show()
```

=====

EDA: INTERCEPTIONS (Int) UNDER-COVERAGE DIAGNOSTIC (DEF)

=====

--- Log-ratio distribution: training bridge vs test bridge ---

Stat	Train mean	Train std	Test mean	Test std
TklW	-0.0029	0.4303	0.1109	0.3017
Int	-0.0289	0.3739	-0.1082	0.4794
Clr	0.0490	0.3628	0.2106	0.4133
Blocks	0.0540	0.4038	-0.0824	0.3374

--- Int detailed distribution ---

Training Bridge (n=106):

Mean=-0.0289 Std=0.3739 Median=-0.0363 IQR=[-0.241, 0.226] Min/Max=[-0.997, 0.855]

Skew=-0.0638 Kurt=0.2092

Test Bridge (n=35):

Mean=-0.1082 Std=0.4794 Median=-0.0993 IQR=[-0.451, 0.182] Min/Max=[-0.928, 1.110]

Skew=0.3799 Kurt=-0.0335

--- Per-league Int log-ratio means (training bridge) ---

Bundesliga	n= 12	mean=-0.0923	std=0.4942
La Liga	n= 18	mean=0.0101	std=0.3451
Ligue 1	n= 25	mean=0.0114	std=0.3649
Premier League	n= 35	mean=-0.0457	std=0.3051
Serie A	n= 16	mean=-0.0514	std=0.4350

--- Prediction interval widths vs true spread ---

Stat	Mean width	True std	Width/2 / std	Coverage
Int	1.0339	0.4794	1.08x	0.629

--- Int per-observation detail ---

Player	Cohort	True	Pred	Lo	Hi	In?
Axel Disasi	2022/23→2023/24	-0.8353	-0.0213	-0.5446	0.4943	✗
Pau Torres	2022/23→2023/24	0.2594	-0.0075	-0.4821	0.5352	✓
Santiago Bueno	2022/23→2023/24	-0.3235	-0.0233	-0.5233	0.5216	✓
Micky van de Ven	2022/23→2023/24	-0.1207	-0.0216	-0.5500	0.5026	✓
Destiny Udogie	2022/23→2023/24	0.1448	-0.0616	-0.5836	0.4600	✓
Nuno Tavares	2022/23→2023/24	-0.9282	-0.0396	-0.5909	0.4664	✗
Konstantinos Mavropanos	2022/23→2023/24	-0.5707	0.0001	-0.4909	0.5352	✗
Joško Gvardiol	2022/23→2023/24	-0.2889	-0.0292	-0.5600	0.4428	✓
Igor	2022/23→2023/24	-0.3332	-0.0375	-0.5637	0.4701	✓
Yasser Larouci	2022/23→2023/24	-0.0895	-0.0296	-0.5526	0.4710	✓
Maxime Estève	2022/23→2023/24	-0.0014	-0.0446	-0.5863	0.4502	✓
Malo Gusto	2022/23→2023/24	-0.5678	-0.0377	-0.5243	0.5022	✗
Gonzalo Montiel	2022/23→2023/24	-0.0993	-0.0115	-0.5504	0.4712	✓
Jakub Kiwior	2022/23→2023/24	-0.6456	-0.0425	-0.5499	0.4731	✗
Álex Moreno	2022/23→2023/24	0.0013	0.0010	-0.5169	0.5081	✓
Issa Kaboré	2022/23→2023/24	-0.2204	-0.0361	-0.5371	0.4854	✓
Ola Aina	2022/23→2023/24	-0.1582	-0.0281	-0.5142	0.4920	✓
Caleb Okoli	2023/24→2024/25	0.2790	-0.0451	-0.6017	0.4460	✓
Sepp van den Berg	2023/24→2024/25	-0.5921	-0.0197	-0.5308	0.4760	✗
Leny Yoro	2023/24→2024/25	0.1356	-0.0495	-0.5659	0.4971	✓
Jean-Clair Todibo	2023/24→2024/25	-0.0070	-0.0322	-0.5076	0.5254	✓
Emmanuel Agbadou	2023/24→2024/25	-0.8767	-0.0226	-0.5356	0.4972	✗

Nikola Milenković	2023/24→2024/25	-0.2668	-0.0474	-0.5365	0.4899	✓
Jake O'Brien	2023/24→2024/25	-0.5966	-0.0356	-0.5807	0.4650	✗
Maxence Lacroix	2023/24→2024/25	0.5128	-0.0137	-0.5472	0.4938	✗
Riccardo Calafiori	2023/24→2024/25	-0.8086	-0.0561	-0.5634	0.4703	✗
Victor Bernth Kristiansen	2023/24→2024/25	1.1097	-0.0490	-0.5445	0.4638	✗
Michael Kayode	2023/24→2024/25	0.3900	-0.0445	-0.5842	0.5007	✓
Radu Drăgușin	2023/24→2024/25	-0.0229	-0.0475	-0.5619	0.4823	✓
Julian Araujo	2023/24→2024/25	0.5389	-0.0317	-0.5085	0.5038	✗
Matthijs de Ligt	2023/24→2024/25	0.1234	-0.0208	-0.5462	0.5114	✓
Ian Maatsen	2023/24→2024/25	0.2326	-0.0458	-0.5912	0.4549	✓
Noussair Mazraoui	2023/24→2024/25	-0.3190	-0.0116	-0.5349	0.5010	✓
Djed Spence	2023/24→2024/25	0.2187	-0.0405	-0.5444	0.4920	✓
Dean Huijsen	2023/24→2024/25	0.9402	-0.0633	-0.5950	0.4337	✗

--- Coverage breakdown by cohort (Int) ---

Int:

2022/23→2023/24: 0.706 (12/17)

2023/24→2024/25: 0.556 (10/18)



```
In [ ]: # =====
# CELL 14: TRANSFER DATA EXPLORATION
# =====

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Transfer Data Exploration', fontsize=14, fontweight='bold')

ax = axes[0, 0]
if df_bridge_att_mid is not None:
    league_counts = df_bridge_att_mid['League_origin'].value_counts()
    ax.barh(league_counts.index, league_counts.values, color='steelblue', alpha=0.7)
    ax.set_xlabel('Count')
    ax.set_title('ATT+MID Transfers by Source League')
    ax.grid(alpha=0.3, axis='x')

ax = axes[0, 1]
if df_bridge_def is not None:
    league_counts = df_bridge_def['League_origin'].value_counts()
    ax.barh(league_counts.index, league_counts.values, color='coral', alpha=0.7)
    ax.set_xlabel('Count')
```

```

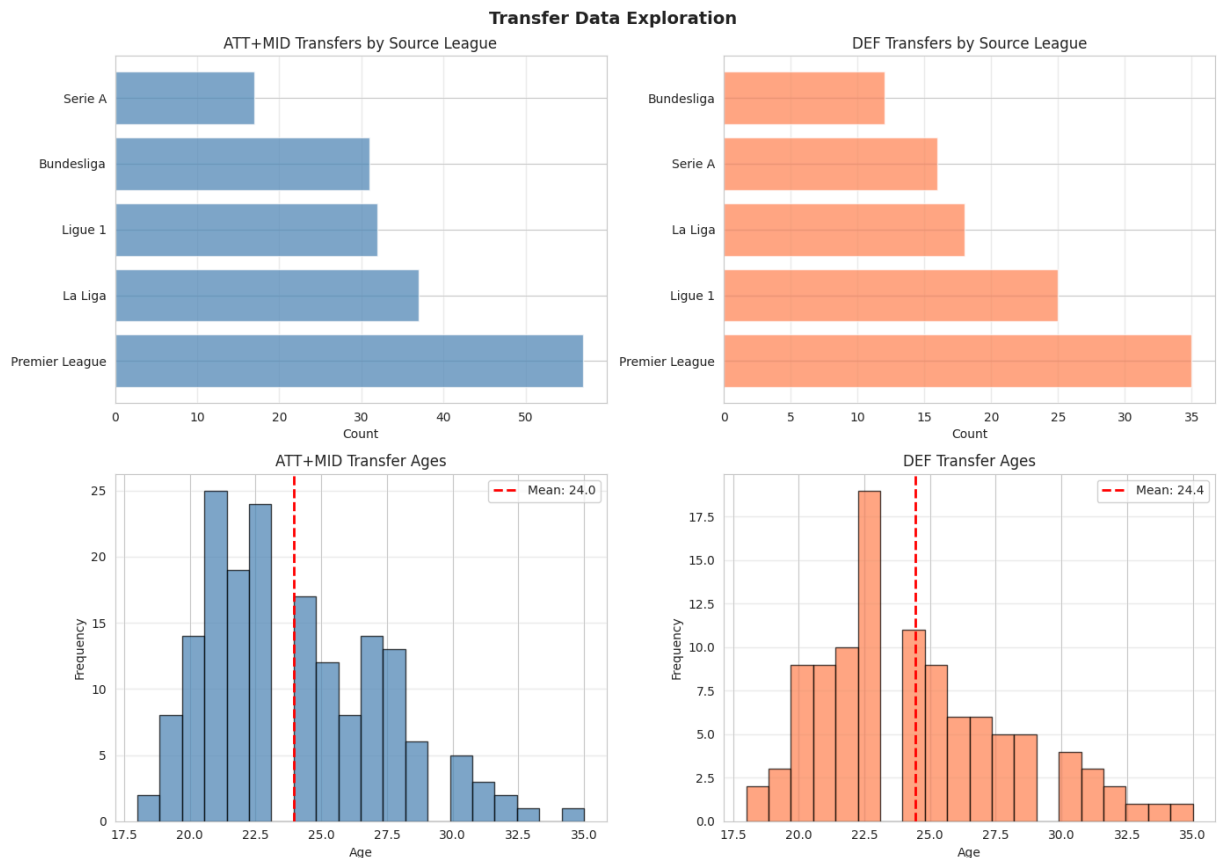
ax.set_title('DEF Transfers by Source League')
ax.grid(alpha=0.3, axis='x')

ax = axes[1, 0]
if df_bridge_att_mid is not None:
    ax.hist(df_bridge_att_mid['Age'], bins=20, color='steelblue', alpha=0.7, edgecolor='black')
    ax.axvline(df_bridge_att_mid['Age'].mean(), color='red', linestyle='--', linewidth=2,
               label=f"Mean: {df_bridge_att_mid['Age'].mean():.1f}")
    ax.set_xlabel('Age')
    ax.set_ylabel('Frequency')
    ax.set_title('ATT+MID Transfer Ages')
    ax.legend()
    ax.grid(alpha=0.3, axis='y')

ax = axes[1, 1]
if df_bridge_def is not None:
    ax.hist(df_bridge_def['Age'], bins=20, color='coral', alpha=0.7, edgecolor='black')
    ax.axvline(df_bridge_def['Age'].mean(), color='red', linestyle='--', linewidth=2,
               label=f"Mean: {df_bridge_def['Age'].mean():.1f}")
    ax.set_xlabel('Age')
    ax.set_ylabel('Frequency')
    ax.set_title('DEF Transfer Ages')
    ax.legend()
    ax.grid(alpha=0.3, axis='y')

plt.tight_layout()
plt.show()

```



```

In [ ]: # =====
# CELL 14b: FIGURE 1 – BRIDGE COMPOSITION (PUBLICATION)

```

```

# =====

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import numpy as np

PUB_STYLE = {
    'font.family':      'serif',
    'font.size':        10,
    'axes.titlesize':   11,
    'axes.labelsize':   10,
    'xtick.labelsize':  9,
    'ytick.labelsize':  9,
    'legend.fontsize':  9,
    'figure.dpi':       300,
    'axes.spines.top':   False,
    'axes.spines.right': False,
    'axes.grid':         True,
    'grid.alpha':        0.3,
    'grid.linestyle':    '--',
}

plt.rcParams.update(PUB_STYLE)

LEAGUE_COLORS = {
    'Ligue 1':      '#4878CF',
    'Bundesliga':   '#6ACC65',
    'Serie A':      '#D65F5F',
    'La Liga':      '#B47CC7',
    'Premier League': '#888888',
}

ATT_COLOR = '#2166AC'
DEF_COLOR = '#B2182B'

fig, axes = plt.subplots(1, 3, figsize=(13, 4.5))
fig.suptitle(
    'Figure 1: Training Bridge Composition',
    fontsize=12, fontweight='bold', y=1.02
)

# Panel A: League distribution (both bridges)
ax = axes[0]
leagues_order = ['Ligue 1', 'Bundesliga', 'Serie A', 'La Liga']
att_counts = [
    len(df_bridge_att_mid[df_bridge_att_mid['League_origin'] == l])
    for l in leagues_order
]
def_counts = [
    len(df_bridge_def[df_bridge_def['League_origin'] == l])
    for l in leagues_order
]
x = np.arange(len(leagues_order))
w = 0.35
bars1 = ax.bar(x - w/2, att_counts, w, label='ATT+MID', color=ATT_COLOR,
               alpha=0.85, edgecolor='white', linewidth=0.8)
bars2 = ax.bar(x + w/2, def_counts, w, label='DEF', color=DEF_COLOR,
               alpha=0.85, edgecolor='white', linewidth=0.8)

```

```

for bar in list(bars1) + list(bars2):
    h = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2, h + 0.4, str(int(h)),
            ha='center', va='bottom', fontsize=8)
ax.set_xticks(x)
ax.set_xticklabels(['Ligue 1', 'Bundesl.', 'Serie A', 'La Liga'])
ax.set_ylabel('Number of Transfers')
ax.set_title('(A) Transfers by Source League')
ax.legend(loc='upper right')
ax.set_ylim(0, max(att_counts + def_counts) * 1.25)

# Panel B: Age distribution
ax = axes[1]
bins = np.arange(17, 38, 1)
ax.hist(df_bridge_att_mid['Age'], bins=bins, color=ATT_COLOR,
        alpha=0.7, edgecolor='white', linewidth=0.6, label='ATT+MID')
ax.hist(df_bridge_def['Age'], bins=bins, color=DEF_COLOR,
        alpha=0.7, edgecolor='white', linewidth=0.6, label='DEF')
ax.axvline(df_bridge_att_mid['Age'].mean(), color=ATT_COLOR, linestyle='--',
           linewidth=1.8, label=f"ATT+MID mean ({df_bridge_att_mid['Age'].mean():.1f})")
ax.axvline(df_bridge_def['Age'].mean(), color=DEF_COLOR, linestyle='--',
           linewidth=1.8, label=f"DEF mean ({df_bridge_def['Age'].mean():.1f})")
ax.set_xlabel('Age at Transfer')
ax.set_ylabel('Frequency')
ax.set_title('(B) Age Distribution at Transfer')
ax.legend(loc='upper right', fontsize=8)

# Panel C: Source minutes (90s) distribution
ax = axes[2]
att_90s = df_bridge_att_mid['90s_src'].values
def_90s = df_bridge_def['90s_src'].values
bins_90 = np.arange(5, 42, 2)
ax.hist(att_90s, bins=bins_90, color=ATT_COLOR,
        alpha=0.7, edgecolor='white', linewidth=0.6, label='ATT+MID')
ax.hist(def_90s, bins=bins_90, color=DEF_COLOR,
        alpha=0.7, edgecolor='white', linewidth=0.6, label='DEF')
ax.axvline(att_90s.mean(), color=ATT_COLOR, linestyle='--',
           linewidth=1.8, label=f"ATT+MID mean ({att_90s.mean():.1f})")
ax.axvline(def_90s.mean(), color=DEF_COLOR, linestyle='--',
           linewidth=1.8, label=f"DEF mean ({def_90s.mean():.1f})")
ax.set_xlabel('Source Season 90-min Appearances')
ax.set_ylabel('Frequency')
ax.set_title('(C) Source Season Playing Time')
ax.legend(loc='upper right', fontsize=8)

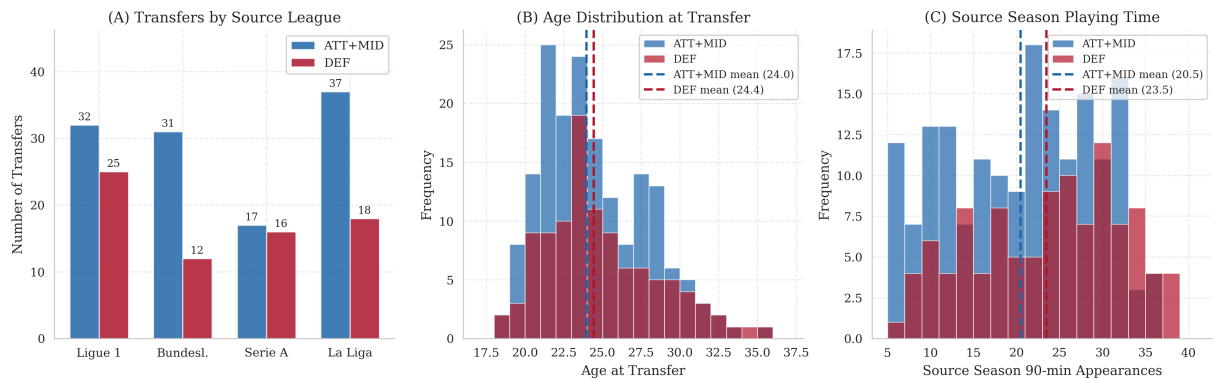
# Caption
n_att = len(df_bridge_att_mid)
n_def = len(df_bridge_def)
pairs = len(df_bridge_att_mid['Transfer_Season'].unique())
fig.text(
    0.5, -0.04,
    f'Training bridge: ATT+MID n={n_att}, DEF n={n_def}, across {pairs} season pairs (2017/'
    ha='center', fontsize=9, style='italic'
)

plt.tight_layout()

```

```
plt.savefig('figure1_bridge_composition.pdf', bbox_inches='tight', dpi=300)
plt.savefig('figure1_bridge_composition.png', bbox_inches='tight', dpi=300)
plt.show()
print("✓ Figure 1 saved")
```

Figure 1: Training Bridge Composition



Training bridge: ATT+MID n=174, DEF n=106, across 5 season pairs (2017/18-2021/22).

✓ Figure 1 saved

```
In [ ]: # =====
# CELL 15: ASSEMBLE RESULTS DICTIONARY
# =====

print("\n" + "=" * 80)
print("ASSEMBLING RESULTS DICTIONARY")
print("=" * 80 + "\n")

all_results = {
    'joint_att_mid': {
        'stats': COMMON_STATS_ATT_MID,
        'joint_trace': joint_trace_att_mid,
        'standardization': standardization_att_mid,
    },
    'joint_def': {
        'stats': COMMON_STATS_DEF,
        'joint_trace': joint_trace_def,
        'standardization': standardization_def,
    },
}

=====
ASSEMBLING RESULTS DICTIONARY
=====
```

```
In [ ]: # =====
# CELL 16: LEAGUE TRANSLATION FUNCTION
# =====

def translate_league(stat_name, src_league, trg_league, results_dict):
    source_key = 'joint_att_mid' if stat_name in COMMON_STATS_ATT_MID else 'joint_def'
    src = results_dict[source_key]
    standardization = src['standardization']
    joint_trace = src['joint_trace']
    stats_list = COMMON_STATS_ATT_MID if stat_name in COMMON_STATS_ATT_MID else COMMON_ST
```



```

stat_idx    = stats_list.index(stat_name)
std         = standardization[stat_name]['std']

def league_tax_mean(league_name):
    if league_name == 'Premier League':
        return 0.0
    l_idx = LEAGUE_MAP[league_name]
    return joint_trace.posterior['league_tax'][:, :, l_idx, stat_idx].mean().item()

src_tax = league_tax_mean(src_league)
trg_tax = league_tax_mean(trg_league)
return float(np.exp((src_tax - trg_tax) * std))

```

```

In [ ]: # =====
# CELL 17: LEAGUE TRANSLATION HEATMAP - ATT+MID
# =====

fig, ax = plt.subplots(figsize=(12, 6))

leagues    = ['Ligue 1', 'Bundesliga', 'Serie A', 'La Liga']
stats_plot = COMMON_STATS_ATT_MID
trans_matrix = []

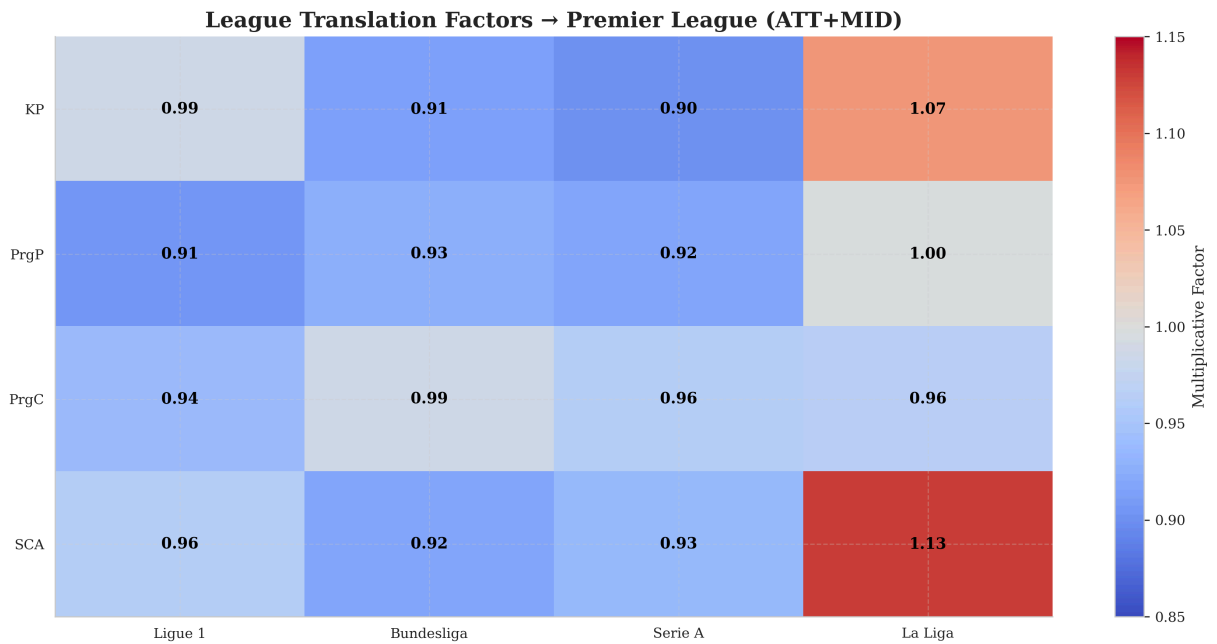
for stat in stats_plot:
    row = []
    for league in leagues:
        factor = translate_league(stat, league, 'Premier League', all_results)
        row.append(factor if factor else 1.0)
    trans_matrix.append(row)

im = ax.imshow(trans_matrix, cmap='coolwarm', aspect='auto', vmin=0.85, vmax=1.15)
ax.set_xticks(range(len(leagues)))
ax.set_yticks(range(len(stats_plot)))
ax.set_xticklabels(leagues)
ax.set_yticklabels(stats_plot)
ax.set_title('League Translation Factors → Premier League (ATT+MID)', fontsize=14, fontwei

for i in range(len(stats_plot)):
    for j in range(len(leagues)):
        ax.text(j, i, f'{trans_matrix[i][j]:.2f}',
                ha='center', va='center', color='black', fontweight='bold')

plt.colorbar(im, ax=ax, label='Multiplicative Factor')
plt.tight_layout()
plt.show()

```



```
In [ ]: # =====
# CELL 17b: FIGURE 4 – LEAGUE TRANSLATION HEATMAP ATT+MID + DEF (PUBLICATION)
# =====

plt.rcParams.update(PUB_STYLE)

leagues_hm = ['Ligue 1', 'Bundesliga', 'Serie A', 'La Liga']

def build_translation_matrix(stats_list):
    matrix = []
    for stat in stats_list:
        row = []
        for league in leagues_hm:
            factor = translate_league(stat, league, 'Premier League', all_results)
            row.append(factor if factor else 1.0)
        matrix.append(row)
    return np.array(matrix)

mat_att = build_translation_matrix(COMMON_STATS_ATT_MID)
mat_def = build_translation_matrix(COMMON_STATS_DEF)

# Shared colour scale centred at 1.0
all_vals = np.concatenate([mat_att.flatten(), mat_def.flatten()])
vmin_hm = min(0.88, all_vals.min() - 0.01)
vmax_hm = max(1.12, all_vals.max() + 0.01)

fig, axes = plt.subplots(1, 2, figsize=(12, 4.5),
                        gridspec_kw={'width_ratios': [4, 4]})
fig.suptitle(
    'Figure 4: League-to-Premier League Translation Factors',
    fontsize=12, fontweight='bold', y=1.02
)

for ax, matrix, stats_list, panel_label, bridge_label in [
    (axes[0], mat_att, COMMON_STATS_ATT_MID, 'A', 'ATT+MID'),
    (axes[1], mat_def, COMMON_STATS_DEF, 'B', 'DEF'),
```

```

]:
    im = ax.imshow(matrix, cmap='RdBu', aspect='auto',
                    vmin=vmin_hm, vmax=vmax_hm)

    ax.set_xticks(range(len(leagues_hm)))
    ax.set_yticks(range(len(stats_list)))
    ax.set_xticklabels(leagues_hm, rotation=30, ha='right', fontsize=9)
    ax.set_yticklabels(stats_list, fontsize=10)
    ax.set_title(f'({panel_label}) {bridge_label}', fontsize=11, fontweight='bold')

    for i in range(len(stats_list)):
        for j in range(len(leagues_hm)):
            val = matrix[i, j]
            text_col = 'white' if abs(val - 1.0) > 0.06 else 'black'
            ax.text(j, i, f'{val:.3f}',
                    ha='center', va='center',
                    fontsize=10, fontweight='bold', color=text_col)

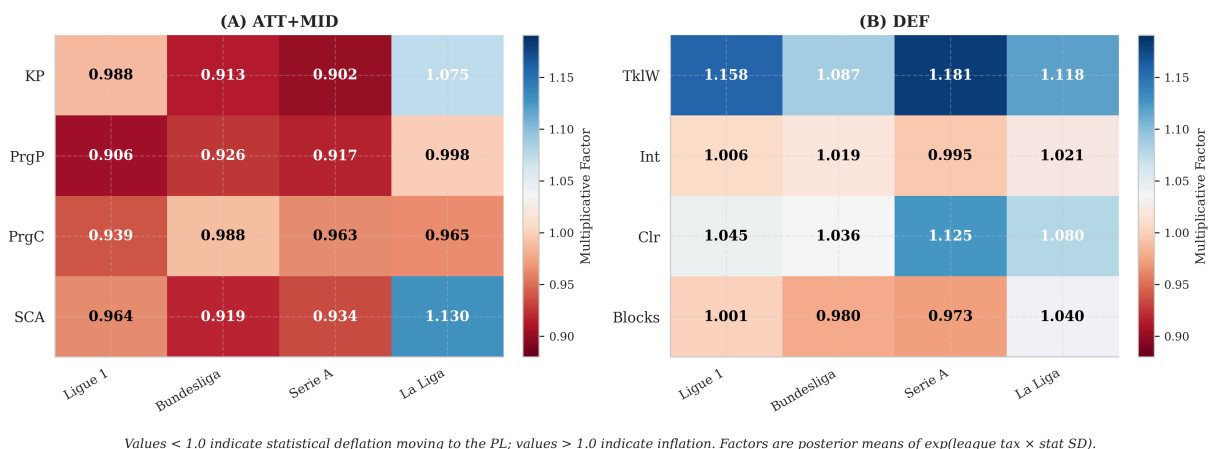
    cb = plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
    cb.set_label('Multiplicative Factor', fontsize=9)
    cb.ax.tick_params(labelsize=8)

    fig.text(
        0.5, -0.04,
        'Values < 1.0 indicate statistical deflation moving to the PL; '
        'values > 1.0 indicate inflation. '
        'Factors are posterior means of exp(league tax × stat SD).',
        ha='center', fontsize=9, style='italic'
    )

    plt.tight_layout()
    plt.savefig('figure4_heatmap.pdf', bbox_inches='tight', dpi=300)
    plt.savefig('figure4_heatmap.png', bbox_inches='tight', dpi=300)
    plt.show()
    print("✓ Figure 4 saved")

```

Figure 4: League-to-Premier League Translation Factors



✓ Figure 4 saved

```

In [ ]: # =====
# CELL 18: ML BASELINES + COMBINED VALIDATION TABLE & FIGURE
# =====

```

```

from sklearn.linear_model import Ridge
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.preprocessing import StandardScaler
from sklearn.multioutput import MultiOutputRegressor
from sklearn.model_selection import cross_val_score, KFold

print("\n" + "=" * 80)
print("ML BASELINES – RIDGE & GBM")
print("=" * 80 + "\n")

def build_ml_features_ridge(df, stats_list, pos_labels=None):
    """Lean contextual feature set for Ridge (avoids source-stat collinearity)."""
    if pos_labels is not None:
        pos_map = {label: idx for idx, label in enumerate(pos_labels)}
        pos_enc = df['Position'].map(pos_map).fillna(0).astype(int).values
    else:
        pos_enc = df['pos_id'].values
    return np.column_stack([
        df['League_ID'].values,
        df['is_foreign'].values,
        df['Age'].values,
        pos_enc,
        df['90s_src'].values,
        df['Age'].values * df['League_ID'].values,
        df['Age'].values * df['is_foreign'].values,
    ])

def build_ml_features_gbm(df, stats_list, pos_labels=None):
    """Full feature set for GBM – includes source stat rates."""
    if pos_labels is not None:
        pos_map = {label: idx for idx, label in enumerate(pos_labels)}
        pos_enc = df['Position'].map(pos_map).fillna(0).astype(int).values
    else:
        pos_enc = df['pos_id'].values
    base = np.column_stack([
        df['League_ID'].values,
        df['is_foreign'].values,
        df['Age'].values,
        pos_enc,
        df['90s_src'].values,
    ])
    src_stats = np.column_stack([df[f'{s}_src'].values for s in stats_list])
    interactions = np.column_stack([
        df['Age'].values * df['League_ID'].values,
        df['Age'].values * df['is_foreign'].values,
    ])
    return np.column_stack([base, src_stats, interactions])

def tune_ridge(X_train_sc, Y_train):
    """
    Independently CV-tune Ridge alpha for this bridge's training split.
    X_train_sc must already be scaled.

```

```

"""
alphas      = [0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0, 50.0, 100.0]
kf          = KFold(n_splits=5, shuffle=True, random_state=42)
best_alpha  = 1.0
best_score  = -np.inf
for alpha in alphas:
    ridge = MultiOutputRegressor(Ridge(alpha=alpha))
    scores = cross_val_score(
        ridge, X_train_sc, Y_train,
        cv=kf, scoring='neg_mean_absolute_error'
    )
    if scores.mean() > best_score:
        best_score = scores.mean()
        best_alpha = alpha
return best_alpha

def tune_gbm(X_train, Y_train):
    """
    CV-tune GBM. max_depth capped at 3 and min_samples_leaf >= 5
    to control variance on the relatively small training set.
    """
    param_grid = [
        {'n_estimators': 100, 'max_depth': 2, 'learning_rate': 0.05, 'min_samples_leaf': 5},
        {'n_estimators': 100, 'max_depth': 3, 'learning_rate': 0.05, 'min_samples_leaf': 5},
        {'n_estimators': 100, 'max_depth': 3, 'learning_rate': 0.10, 'min_samples_leaf': 5},
        {'n_estimators': 200, 'max_depth': 2, 'learning_rate': 0.05, 'min_samples_leaf': 5},
        {'n_estimators': 200, 'max_depth': 3, 'learning_rate': 0.05, 'min_samples_leaf': 5}
    ]
    kf          = KFold(n_splits=5, shuffle=True, random_state=42)
    best_params = param_grid[0]
    best_score  = -np.inf
    for params in param_grid:
        gbm = MultiOutputRegressor(
            GradientBoostingRegressor(
                n_estimators = params['n_estimators'],
                max_depth     = params['max_depth'],
                learning_rate = params['learning_rate'],
                min_samples_leaf = params['min_samples_leaf'],
                random_state  = 42,
            )
        )
        scores = cross_val_score(
            gbm, X_train, Y_train,
            cv=kf, scoring='neg_mean_absolute_error'
        )
        if scores.mean() > best_score:
            best_score = scores.mean()
            best_params = params
    return best_params

def compute_cv_prediction_intervals(model, X_train, Y_train, X_test, n_splits=5):
    """
    Out-of-fold residual std for prediction intervals (not training residuals).
    Refits on full training set before predicting on test.

```

```

"""
kf          = KFold(n_splits=n_splits, shuffle=True, random_state=42)
oof_residuals = []
for train_idx, val_idx in kf.split(X_train):
    X_tr, X_val = X_train[train_idx], X_train[val_idx]
    Y_tr, Y_val = Y_train[train_idx], Y_train[val_idx]
    model.fit(X_tr, Y_tr)
    oof_residuals.append(Y_val - model.predict(X_val))
oof_residuals = np.vstack(oof_residuals)
resid_std     = oof_residuals.std(axis=0, ddof=1)
model.fit(X_train, Y_train)
Y_pred_test = model.predict(X_test)
return Y_pred_test, Y_pred_test - 1.64 * resid_std, Y_pred_test + 1.64 * resid_std

def run_ml_baselines(df_train, Y_train_std,
                    df_test, Y_test_std,
                    stats_list, standardization,
                    pos_labels_train, label):

    X_train_ridge = build_ml_features_ridge(df_train, stats_list)
    X_test_ridge  = build_ml_features_ridge(df_test, stats_list,
                                           pos_labels=pos_labels_train)

    scaler        = StandardScaler()
    X_train_ridge_sc = scaler.fit_transform(X_train_ridge)
    X_test_ridge_sc  = scaler.transform(X_test_ridge)

    X_train_gbm = build_ml_features_gbm(df_train, stats_list)
    X_test_gbm  = build_ml_features_gbm(df_test, stats_list,
                                       pos_labels=pos_labels_train)

    print(f" Tuning Ridge ({label})...")
    best_alpha = tune_ridge(X_train_ridge_sc, Y_train_std)
    print(f" Best Ridge alpha: {best_alpha}")
    ridge = MultiOutputRegressor(Ridge(alpha=best_alpha))

    print(f" Tuning GBM ({label})...")
    best_params = tune_gbm(X_train_gbm, Y_train_std)
    print(f" Best GBM params: {best_params}")
    gbm = MultiOutputRegressor(
        GradientBoostingRegressor(
            n_estimators = best_params['n_estimators'],
            max_depth    = best_params['max_depth'],
            learning_rate = best_params['learning_rate'],
            min_samples_leaf= best_params['min_samples_leaf'],
            random_state  = 42,
        )
    )

    print(f" Computing CV prediction intervals ({label})...")
    Y_pred_ridge, ridge_p5, ridge_p95 = compute_cv_prediction_intervals(
        ridge, X_train_ridge_sc, Y_train_std, X_test_ridge_sc
    )
    Y_pred_gbm, gbm_p5, gbm_p95 = compute_cv_prediction_intervals(
        gbm, X_train_gbm, Y_train_std, X_test_gbm
    )

```

```

results = []
for j, stat in enumerate(stats_list):
    mean_s = standardization[stat]['mean']
    std_s = standardization[stat]['std']
    y_true = Y_test_std[:, j] * std_s + mean_s
    y_pred_r = Y_pred_ridge[:, j] * std_s + mean_s
    y_pred_g = Y_pred_gbm[:, j] * std_s + mean_s
    y_lo_r = ridge_p5[:, j] * std_s + mean_s
    y_hi_r = ridge_p95[:, j] * std_s + mean_s
    y_lo_g = gbm_p5[:, j] * std_s + mean_s
    y_hi_g = gbm_p95[:, j] * std_s + mean_s
    results.append({
        'Stat': stat,
        'Model': 'Baseline: Ridge',
        'MAE': round(float(np.mean(np.abs(y_true - y_pred_r))), 4),
        'RMSE': round(float(np.sqrt(np.mean((y_true - y_pred_r) ** 2))), 4),
        'Coverage_90': round(float(np.mean((y_true >= y_lo_r) & (y_true <= y_hi_r))), 3
        'N_test': len(df_test),
    })
    results.append({
        'Stat': stat,
        'Model': 'Baseline: GBM',
        'MAE': round(float(np.mean(np.abs(y_true - y_pred_g))), 4),
        'RMSE': round(float(np.sqrt(np.mean((y_true - y_pred_g) ** 2))), 4),
        'Coverage_90': round(float(np.mean((y_true >= y_lo_g) & (y_true <= y_hi_g))), 3
        'N_test': len(df_test),
    })
return pd.DataFrame(results)

if not df_test_att.empty:
    ml_att = run_ml_baselines(
        df_bridge_att_mid, Y_std_att_mid,
        df_test_att, Y_test_att,
        COMMON_STATS_ATT_MID, standardization_att_mid,
        pos_labels_att_mid, label='ATT+MID',
    )
    print("\nML Baselines - ATT+MID:")
    print(ml_att.to_string(index=False))
    print()

if not df_test_def.empty:
    ml_def = run_ml_baselines(
        df_bridge_def, Y_std_def,
        df_test_def, Y_test_def,
        COMMON_STATS_DEF, standardization_def,
        pos_labels_def, label='DEF',
    )
    print("ML Baselines - DEF:")
    print(ml_def.to_string(index=False))
    print()

# --- Combined validation table & figure ---
if not df_test_att.empty and not df_test_def.empty:
    val_all_att = pd.concat([val_hb_att, val_uni_att, ml_att], ignore_index=True)

```

```

val_all_def = pd.concat([val_hb_def, val_uni_def, ml_def], ignore_index=True)

print("=" * 80)
print("FULL VALIDATION TABLE – ATT+MID")
print("=" * 80)
print(val_all_att[['Stat', 'Model', 'MAE', 'RMSE', 'Coverage_90']].to_string(index=False))

print()
print("=" * 80)
print("FULL VALIDATION TABLE – DEF")
print("=" * 80)
print(val_all_def[['Stat', 'Model', 'MAE', 'RMSE', 'Coverage_90']].to_string(index=False))

model_colors = {
    'HBFLT (Hierarchical)': 'steelblue',
    'Baseline: Univariate': 'coral',
    'Baseline: Ridge': 'seagreen',
    'Baseline: GBM': 'goldenrod',
}

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
fig.suptitle(
    'Out-of-Sample Validation (2023/2024 & 2024/25 held-out): HBFLT vs Baselines',
    fontsize=14, fontweight='bold'
)
for ax, val_df, title in [
    (axes[0], val_all_att, f'ATT+MID (n={len(df_test_att)} test transfers)'),
    (axes[1], val_all_def, f'DEF (n={len(df_test_def)} test transfers)'),
]:
    pivot = val_df.pivot_table(values='MAE', index='Stat', columns='Model')
    col_order = [c for c in model_colors if c in pivot.columns]
    colors = [model_colors[c] for c in col_order]
    pivot[col_order].plot(kind='bar', ax=ax, color=colors, alpha=0.75, edgecolor='white')
    ax.set_title(title, fontsize=12)
    ax.set_ylabel('MAE (log-ratio scale)', fontsize=11)
    ax.set_xlabel('Statistic', fontsize=11)
    ax.legend(title='Model', fontsize=9, title_fontsize=9)
    ax.grid(alpha=0.3, axis='y')
    ax.tick_params(axis='x', rotation=45)
plt.tight_layout()
plt.show()

```


=====

ML BASELINES – RIDGE & GBM

=====

Tuning Ridge (ATT+MID)...

Best Ridge alpha: 100.0

Tuning GBM (ATT+MID)...

Best GBM params: {'n_estimators': 100, 'max_depth': 2, 'learning_rate': 0.05, 'min_samples_leaf': 5}

Computing CV prediction intervals (ATT+MID)...

ML Baselines – ATT+MID:

Stat	Model	MAE	RMSE	Coverage_90	N_test
KP	Baseline: Ridge	0.3411	0.4557	0.978	45
KP	Baseline: GBM	0.3079	0.3980	0.956	45
PrgP	Baseline: Ridge	0.2920	0.4173	0.911	45
PrgP	Baseline: GBM	0.3079	0.4028	0.867	45
PrgC	Baseline: Ridge	0.3312	0.4189	0.933	45
PrgC	Baseline: GBM	0.3296	0.4253	0.911	45
SCA	Baseline: Ridge	0.2469	0.3238	0.978	45
SCA	Baseline: GBM	0.2613	0.3342	0.956	45

Tuning Ridge (DEF)...

Best Ridge alpha: 100.0

Tuning GBM (DEF)...

Best GBM params: {'n_estimators': 100, 'max_depth': 2, 'learning_rate': 0.05, 'min_samples_leaf': 5}

Computing CV prediction intervals (DEF)...

ML Baselines – DEF:

Stat	Model	MAE	RMSE	Coverage_90	N_test
TklW	Baseline: Ridge	0.2411	0.3035	0.943	35
TklW	Baseline: GBM	0.2483	0.3005	0.971	35
Int	Baseline: Ridge	0.3896	0.4905	0.829	35
Int	Baseline: GBM	0.3891	0.4815	0.800	35
Clr	Baseline: Ridge	0.3276	0.4420	0.914	35
Clr	Baseline: GBM	0.2915	0.3852	0.886	35
Blocks	Baseline: Ridge	0.2840	0.3656	0.914	35
Blocks	Baseline: GBM	0.2549	0.3286	0.943	35

=====

FULL VALIDATION TABLE – ATT+MID

=====

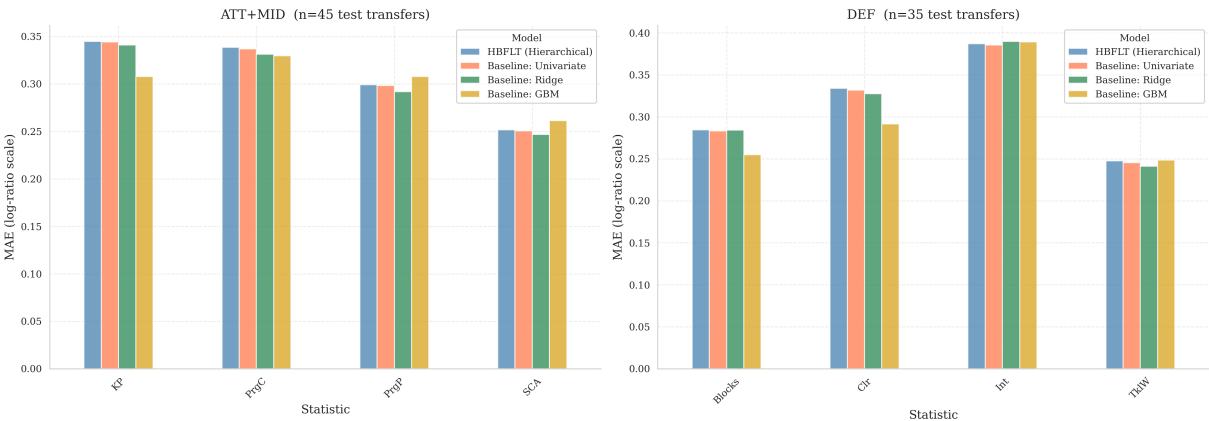
Stat	Model	MAE	RMSE	Coverage_90
KP	HBFLT (Hierarchical)	0.3447	0.4621	0.933
PrgP	HBFLT (Hierarchical)	0.2992	0.4283	0.889
PrgC	HBFLT (Hierarchical)	0.3387	0.4218	0.889
SCA	HBFLT (Hierarchical)	0.2516	0.3293	0.911
KP	Baseline: Univariate	0.3444	0.4629	0.956
PrgP	Baseline: Univariate	0.2983	0.4280	0.889
PrgC	Baseline: Univariate	0.3370	0.4199	0.956
SCA	Baseline: Univariate	0.2505	0.3278	0.956
KP	Baseline: Ridge	0.3411	0.4557	0.978
KP	Baseline: GBM	0.3079	0.3980	0.956
PrgP	Baseline: Ridge	0.2920	0.4173	0.911
PrgP	Baseline: GBM	0.3079	0.4028	0.867
PrgC	Baseline: Ridge	0.3312	0.4189	0.933

PrgC	Baseline: GBM	0.3296	0.4253	0.911
SCA	Baseline: Ridge	0.2469	0.3238	0.978
SCA	Baseline: GBM	0.2613	0.3342	0.956

FULL VALIDATION TABLE – DEF

Stat	Model	MAE	RMSE	Coverage_90
TklW	HBFLT (Hierarchical)	0.2474	0.3074	0.943
Int	HBFLT (Hierarchical)	0.3869	0.4894	0.629
Clr	HBFLT (Hierarchical)	0.3338	0.4415	0.800
Blocks	HBFLT (Hierarchical)	0.2843	0.3664	0.857
TklW	Baseline: Univariate	0.2453	0.3064	0.943
Int	Baseline: Univariate	0.3855	0.4886	0.829
Clr	Baseline: Univariate	0.3318	0.4404	0.886
Blocks	Baseline: Univariate	0.2833	0.3658	0.914
TklW	Baseline: Ridge	0.2411	0.3035	0.943
TklW	Baseline: GBM	0.2483	0.3005	0.971
Int	Baseline: Ridge	0.3896	0.4905	0.829
Int	Baseline: GBM	0.3891	0.4815	0.800
Clr	Baseline: Ridge	0.3276	0.4420	0.914
Clr	Baseline: GBM	0.2915	0.3852	0.886
Blocks	Baseline: Ridge	0.2840	0.3656	0.914
Blocks	Baseline: GBM	0.2549	0.3286	0.943

Out-of-Sample Validation (2023/2024 & 2024/25 held-out): HBFLT vs Baselines



```
In [ ]: # =====
# CELL 18b: FIGURE 5 – OOS VALIDATION (PUBLICATION)
# =====
# Redesigned from grouped bar chart to dot plot with coverage annotations.
# Cleaner for a journal figure with 4 models × 4 stats.

plt.rcParams.update(PUB_STYLE)

MODEL_STYLES = {
    'HBFLT (Hierarchical)': {'color': '#2166AC', 'marker': 'o', 'ms': 8, 'zorder': 5},
    'Baseline: Univariate': {'color': '#777777', 'marker': 's', 'ms': 6, 'zorder': 4},
    'Baseline: Ridge':      {'color': '#D6604D', 'marker': '^', 'ms': 6, 'zorder': 3},
    'Baseline: GBM':        {'color': '#4DAC26', 'marker': 'D', 'ms': 6, 'zorder': 3},
}

NOMINAL_COV = 0.90

fig, axes = plt.subplots(2, 2, figsize=(13, 9))
```

```

fig.suptitle(
    'Figure 5: Out-of-Sample Validation – HBFLT vs Baselines\n'
    f'Held-out test set: ATT+MID n={len(df_test_att)}, DEF n={len(df_test_def)}',
    fontsize=12, fontweight='bold', y=1.01
)

for row_idx, (val_df, stats_list, bridge_label) in enumerate([
    (val_all_att, COMMON_STATS_ATT_MID, 'ATT+MID'),
    (val_all_def, COMMON_STATS_DEF, 'DEF'),
]):
    # Panel left: MAE dot plot
    ax_mae = axes[row_idx, 0]
    ax_cov = axes[row_idx, 1]

    stats_x = {s: i for i, s in enumerate(stats_list)}
    x_vals = np.arange(len(stats_list))
    n_models = len(MODEL_STYLES)
    offsets = np.linspace(-0.28, 0.28, n_models)

    for m_idx, (model_name, style) in enumerate(MODEL_STYLES.items()):
        model_df = val_df[val_df['Model'] == model_name]
        if model_df.empty:
            continue
        mae_vals = []
        cov_vals = []
        xs = []
        for s in stats_list:
            row = model_df[model_df['Stat'] == s]
            if not row.empty:
                mae_vals.append(row['MAE'].values[0])
                cov_vals.append(row['Coverage_90'].values[0])
                xs.append(stats_x[s] + offsets[m_idx])

        ax_mae.scatter(
            xs, mae_vals,
            color=style['color'], marker=style['marker'],
            s=style['ms']**2, zorder=style['zorder'],
            label=model_name, edgecolors='white', linewidths=0.5,
        )
        ax_mae.plot(
            xs, mae_vals,
            color=style['color'], linewidth=0.8, alpha=0.5, zorder=style['zorder'] - 1,
        )

        ax_cov.scatter(
            xs, cov_vals,
            color=style['color'], marker=style['marker'],
            s=style['ms']**2, zorder=style['zorder'],
            label=model_name, edgecolors='white', linewidths=0.5,
        )
        ax_cov.plot(
            xs, cov_vals,
            color=style['color'], linewidth=0.8, alpha=0.5, zorder=style['zorder'] - 1,
        )

    # MAE panel formatting

```

```

ax_mae.set_xticks(x_vals)
ax_mae.set_xticklabels(stats_list)
ax_mae.set_ylabel('MAE (log-ratio scale)')
ax_mae.set_title(f'({chr(65 + row_idx*2)}) {bridge_label} - Point Prediction (MAE)')
ax_mae.set_ylim(bottom=0)
if row_idx == 0:
    ax_mae.legend(loc='upper right', fontsize=8, framealpha=0.9)

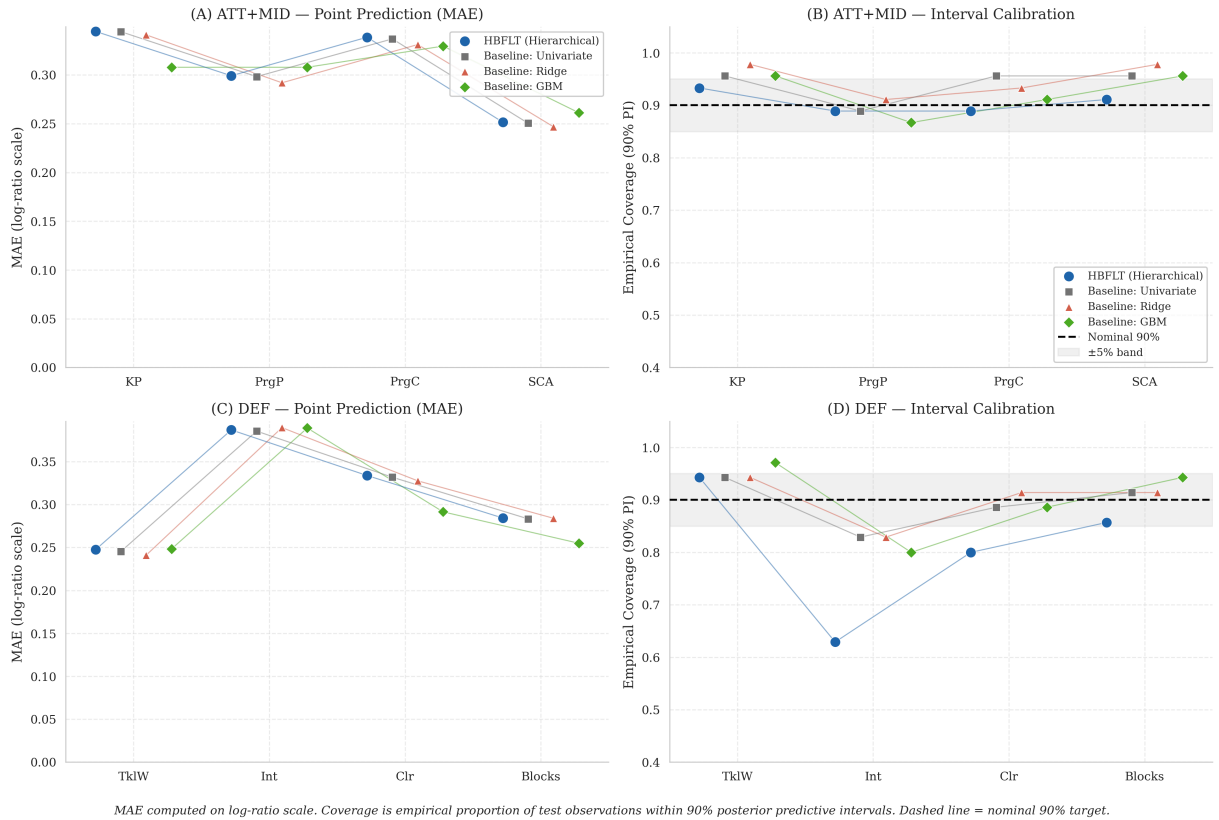
# Coverage panel formatting
ax_cov.axhline(NOMINAL_COV, color='black', linewidth=1.5,
               linestyle='--', label=f'Nominal {NOMINAL_COV:.0%}', zorder=6)
ax_cov.fill_between(
    [-0.5, len(stats_list) - 0.5],
    [NOMINAL_COV - 0.05] * 2, [NOMINAL_COV + 0.05] * 2,
    color='grey', alpha=0.10, label='±5% band',
)
ax_cov.set_xticks(x_vals)
ax_cov.set_xticklabels(stats_list)
ax_cov.set_ylabel('Empirical Coverage (90% PI)')
ax_cov.set_ylim(0.4, 1.05)
ax_cov.set_title(f'({chr(66 + row_idx*2)}) {bridge_label} - Interval Calibration')
ax_cov.set_xlim(-0.5, len(stats_list) - 0.5)
ax_mae.set_xlim(-0.5, len(stats_list) - 0.5)
if row_idx == 0:
    ax_cov.legend(loc='lower right', fontsize=8, framealpha=0.9)

fig.text(
    0.5, -0.02,
    'MAE computed on log-ratio scale. Coverage is empirical proportion of test observations'
    'within 90% posterior predictive intervals. Dashed line = nominal 90% target.',
    ha='center', fontsize=9, style='italic'
)

plt.tight_layout()
plt.savefig('figure5_oos_validation.pdf', bbox_inches='tight', dpi=300)
plt.savefig('figure5_oos_validation.png', bbox_inches='tight', dpi=300)
plt.show()
print("✓ Figure 5 saved")

```

Figure 5: Out-of-Sample Validation — HBFLT vs Baselines
Held-out test set: ATT+MID n=45, DEF n=35



✓ Figure 5 saved

```
In [ ]: # =====
# CELL 19: CALIBRATION CURVES (held-out 2023/2024 & 2024/25 test set)
# =====

print("\n" + "=" * 80)
print("CALIBRATION CURVES (OOS)")
print("=" * 80 + "\n")

def _unpack_chol(chol_packed, n_stats_loc):
    """Reconstruct lower-triangular Cholesky matrices from PyMC's packed form."""
    n_samples = chol_packed.shape[0]
    chol_full = np.zeros((n_samples, n_stats_loc, n_stats_loc))
    idx = np.tril_indices(n_stats_loc)
    for s in range(n_samples):
        chol_full[s][idx] = chol_packed[s]
    return chol_full

def compute_calibration_oos(df_test, Y_test_std, stats_list, standardization,
                           joint_trace, age_mean, label, n_nominal=19):
    """Full posterior predictive calibration across a grid of nominal coverage levels."""
    n_obs = len(df_test)
    n_stats_loc = len(stats_list)

    alpha_samp = joint_trace.posterior['alpha'].values.reshape(-1, n_stats_loc)
    n_samples = alpha_samp.shape[0]
```

```

lt_samp = joint_trace.posterior['league_tax'].values.reshape(-1, 4, n_stats_loc)
ba_samp = joint_trace.posterior['beta_age'].values.reshape(-1, n_stats_loc)
ba2_samp = joint_trace.posterior['beta_age2'].values.reshape(-1, n_stats_loc)

chol_raw = joint_trace.posterior['chol'].values
n_packed = n_stats_loc * (n_stats_loc + 1) // 2
chol_flat = chol_raw.reshape(-1, n_packed)
assert chol_flat.shape[1] == n_packed, (
    f"Chol packed size mismatch: got {chol_flat.shape[1]}, "
    f"expected {n_packed} for n_stats={n_stats_loc}."
)
chol_samp = _unpack_chol(chol_flat, n_stats_loc)

rng = np.random.default_rng(42)
nu = 6

all_preds = np.zeros((n_obs, n_samples, n_stats_loc))
for i in range(n_obs):
    age_c = (df_test.iloc[i]['Age'] - age_mean) / 5.0
    is_for = int(df_test.iloc[i]['is_foreign'])
    l_idx = int(np.clip(df_test.iloc[i]['League_ID'], 0, 3))
    age_eff = ba_samp * age_c + ba2_samp * (age_c ** 2)
    tax = lt_samp[:, l_idx, :] if is_for else np.zeros((n_samples, n_stats_loc))
    mu_samp = alpha_samp + tax + age_eff
    chi2_i = rng.chisquare(nu, size=n_samples) / nu
    z_i = rng.standard_normal(size=(n_samples, n_stats_loc))
    t_i = z_i / np.sqrt(chi2_i[:, None])
    noise_i = np.einsum('sij,sj->si', chol_samp, t_i)
    all_preds[i] = mu_samp + noise_i

nominal_levels = np.linspace(0.05, 0.95, n_nominal)
results = []
for j, stat in enumerate(stats_list):
    mean_s = standardization[stat]['mean']
    std_s = standardization[stat]['std']
    y_true = Y_test_std[:, j] * std_s + mean_s
    for nom in nominal_levels:
        lo_p = (1 - nom) / 2 * 100
        hi_p = (1 + nom) / 2 * 100
        preds_orig = all_preds[:, :, j] * std_s + mean_s
        lo = np.percentile(preds_orig, lo_p, axis=1)
        hi = np.percentile(preds_orig, hi_p, axis=1)
        emp = float(np.mean((y_true >= lo) & (y_true <= hi)))
        results.append({
            'Stat': stat,
            'Nominal': round(float(nom), 2),
            'Empirical': round(emp, 3),
            'Delta': round(emp - float(nom), 3),
        })

df_cal = pd.DataFrame(results)
print(f"{label} - Calibration at 90% nominal:")
print(df_cal[df_cal['Nominal'] == 0.9][['Stat', 'Nominal', 'Empirical', 'Delta']].to_st
return df_cal

```

```

cal_att_oos = None
cal_def_oos = None

if not df_test_att.empty:
    cal_att_oos = compute_calibration_oos(
        df_test_att, Y_test_att,
        COMMON_STATS_ATT_MID, standardization_att_mid,
        joint_trace_att_mid, age_mean_att_mid,
        label='ATT+MID',
    )

if not df_test_def.empty:
    cal_def_oos = compute_calibration_oos(
        df_test_def, Y_test_def,
        COMMON_STATS_DEF, standardization_def,
        joint_trace_def, age_mean_def,
        label='DEF',
    )

if cal_att_oos is not None and cal_def_oos is not None:
    fig, axes = plt.subplots(1, 2, figsize=(14, 6))
    fig.suptitle(
        'Calibration Curves: Nominal vs Empirical Coverage (held-out 2023/24 & 2024/25)',
        fontsize=14, fontweight='bold'
    )
    for ax, cal_df, stats_list_plot, title in [
        (axes[0], cal_att_oos, COMMON_STATS_ATT_MID, 'ATT+MID'),
        (axes[1], cal_def_oos, COMMON_STATS_DEF, 'DEF'),
    ]:
        ax.plot([0, 1], [0, 1], 'k--', linewidth=2, label='Perfect calibration', zorder=5)
        for stat in stats_list_plot:
            sub = cal_df[cal_df['Stat'] == stat]
            ax.plot(
                sub['Nominal'].values, sub['Empirical'].values,
                marker='o', markersize=4, label=stat, alpha=0.85, linewidth=1.5,
            )
        ax.set_xlabel('Nominal Coverage', fontsize=11)
        ax.set_ylabel('Empirical Coverage', fontsize=11)
        ax.set_title(title, fontsize=12)
        ax.legend(fontsize=9)
        ax.grid(alpha=0.3)
        ax.set_xlim(0, 1)
        ax.set_ylim(0, 1)
    plt.tight_layout()
    plt.show()

```

CALIBRATION CURVES (OOS)

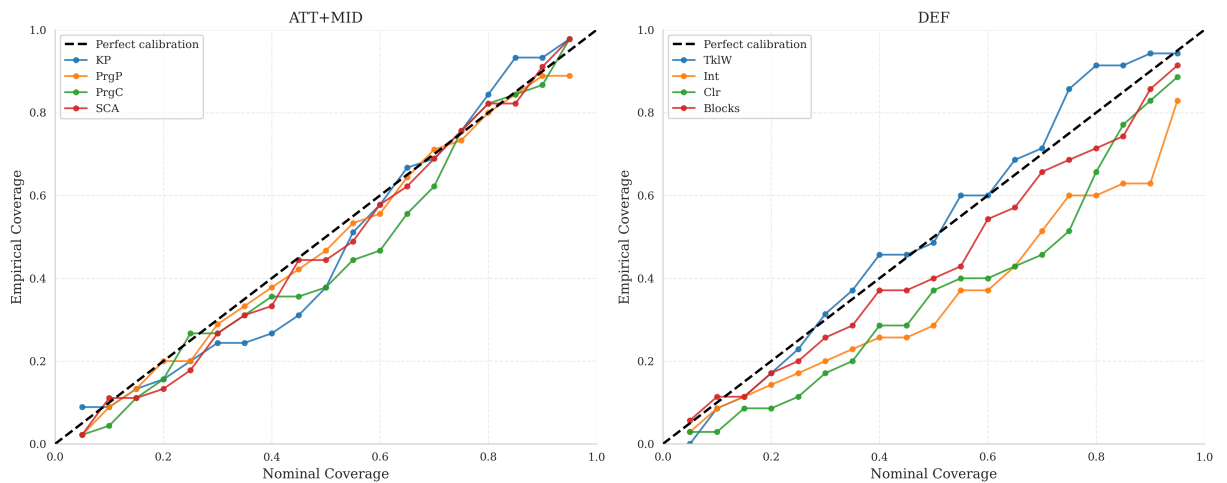
ATT+MID – Calibration at 90% nominal:

Stat	Nominal	Empirical	Delta
KP	0.9	0.933	0.033
PrgP	0.9	0.889	-0.011
PrgC	0.9	0.867	-0.033
SCA	0.9	0.911	0.011

DEF – Calibration at 90% nominal:

Stat	Nominal	Empirical	Delta
TklW	0.9	0.943	0.043
Int	0.9	0.629	-0.271
Clr	0.9	0.829	-0.071
Blocks	0.9	0.857	-0.043

Calibration Curves: Nominal vs Empirical Coverage (held-out 2023/24 & 2024/25)



```
In [ ]: # =====
# CELL 19b: FIGURE 6 – CALIBRATION CURVES (PUBLICATION)
# =====

plt.rcParams.update(PUB_STYLE)

STAT_COLORS_ATT = {
    'KP': '#2166AC',
    'PrgP': '#4DAC26',
    'PrgC': '#D6604D',
    'SCA': '#762A83',
}

STAT_COLORS_DEF = {
    'TklW': '#1B7837',
    'Int': '#D6604D',
    'Clr': '#4393C3',
    'Blocks': '#8C510A',
}

fig, axes = plt.subplots(1, 2, figsize=(11, 5))
fig.suptitle(
    'Figure 6: Calibration Curves – Nominal vs Empirical Coverage\n'
    'HBFLT Posterior Predictive Intervals, Held-out Test Set',
```



```

        fontsize=12, fontweight='bold', y=1.02
    )

    for ax, cal_df, stats_list, stat_colors, panel_label, bridge_label in [
        (axes[0], cal_att_oos, COMMON_STATS_ATT_MID, STAT_COLORS_ATT, 'A', 'ATT+MID'),
        (axes[1], cal_def_oos, COMMON_STATS_DEF, STAT_COLORS_DEF, 'B', 'DEF'),
    ]:
        # Perfect calibration line
        ax.plot([0, 1], [0, 1], 'k--', linewidth=1.8,
                label='Perfect calibration', zorder=10, alpha=0.8)

        # ±5% band
        x_band = np.linspace(0, 1, 100)
        ax.fill_between(x_band, x_band - 0.05, x_band + 0.05,
                       color='grey', alpha=0.12, label='±5% band')

        for stat in stats_list:
            sub = cal_df[cal_df['Stat'] == stat]
            nom = sub['Nominal'].values
            emp = sub['Empirical'].values
            col = stat_colors[stat]
            ax.plot(nom, emp, marker='o', markersize=4.5,
                    linewidth=1.8, color=col, label=stat,
                    alpha=0.9, zorder=5)
            # Annotate 90% nominal point
            idx_90 = np.argmin(np.abs(nom - 0.90))
            ax.annotate(
                f'{emp[idx_90]:.2f}',
                xy=(nom[idx_90], emp[idx_90]),
                xytext=(5, 5), textcoords='offset points',
                fontsize=7.5, color=col,
            )

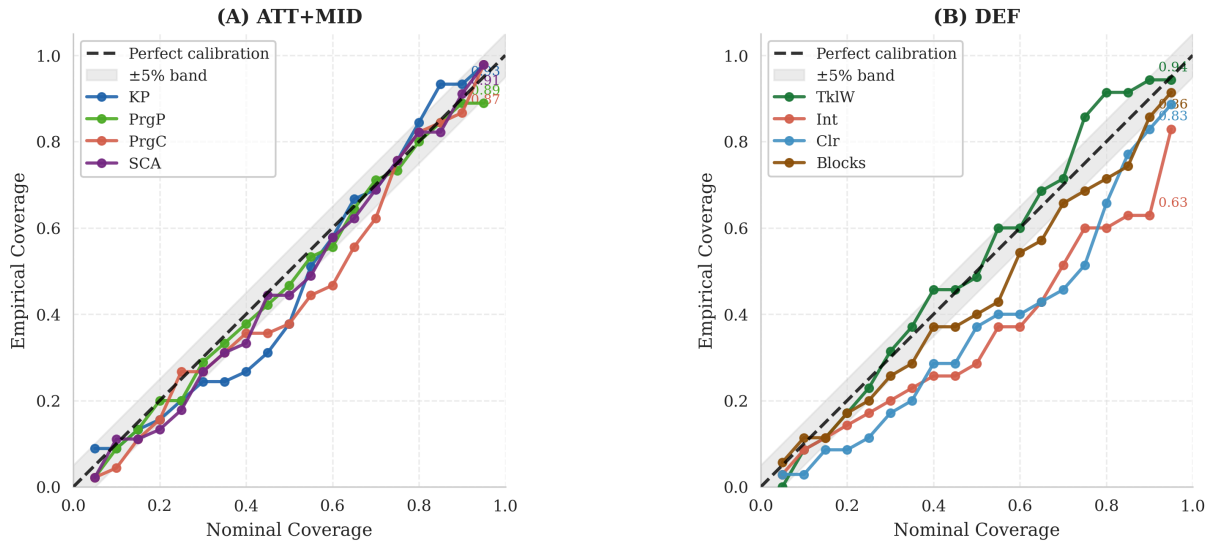
        ax.set_xlabel('Nominal Coverage', fontsize=10)
        ax.set_ylabel('Empirical Coverage', fontsize=10)
        ax.set_title(f'({panel_label}) {bridge_label}', fontsize=11, fontweight='bold')
        ax.set_xlim(0.0, 1.0)
        ax.set_ylim(0.0, 1.05)
        ax.legend(fontsize=8.5, loc='upper left', framealpha=0.9)
        ax.set_aspect('equal')
        ax.grid(True, alpha=0.3, linestyle='--')
        ax.spines['top'].set_visible(False)
        ax.spines['right'].set_visible(False)

    fig.text(
        0.5, -0.04,
        'Annotations show empirical coverage at the 90% nominal level. '
        'Points above the diagonal indicate over-coverage; below indicates under-coverage.',
        ha='center', fontsize=9, style='italic'
    )

    plt.tight_layout()
    plt.savefig('figure6_calibration.pdf', bbox_inches='tight', dpi=300)
    plt.savefig('figure6_calibration.png', bbox_inches='tight', dpi=300)
    plt.show()
    print("✓ Figure 6 saved")

```

Figure 6: Calibration Curves — Nominal vs Empirical Coverage
HBFLT Posterior Predictive Intervals, Held-out Test Set



Annotations show empirical coverage at the 90% nominal level. Points above the diagonal indicate over-coverage; below indicates under-coverage.

✓ Figure 6 saved

```
In [ ]: # =====
# CELL 20: POSTERIOR CREDIBLE INTERVAL TABLE – LEAGUE EFFECTS
# =====

print("\n" + "=" * 80)
print("POSTERIOR CREDIBLE INTERVAL TABLE – LEAGUE EFFECTS")
print("=" * 80 + "\n")

def build_posterior_sd_table(joint_trace, stats_list, label):
    lt = joint_trace.posterior['league_tax'].values
    n_chains, n_draws, n_leagues, n_stats_loc = lt.shape
    flat = lt.reshape(n_chains * n_draws, n_leagues, n_stats_loc)

    leagues = ['Ligue 1', 'Bundesliga', 'Serie A', 'La Liga']
    rows = []
    for l_idx, league in enumerate(leagues):
        for s_idx, stat in enumerate(stats_list):
            samp = flat[:, l_idx, s_idx]
            hdi_vals = az.hdi(samp, hdi_prob=0.90)
            rows.append({
                'League': league,
                'Stat': stat,
                'Post. Mean': round(float(np.mean(samp)), 4),
                'Post. SD': round(float(np.std(samp, ddof=1)), 4),
                'CI 5%': round(float(hdi_vals[0]), 4),
                'CI 95%': round(float(hdi_vals[1]), 4),
            })

    df_sd = pd.DataFrame(rows)
    print(f"{label} – Posterior League Tax Summary (90% HDI):\n")
    print(df_sd.to_string(index=False))
    return df_sd
```

```
sd_table_att = build_posterior_sd_table(
    joint_trace_att_mid, COMMON_STATS_ATT_MID, 'ATT+MID'
)
print()
sd_table_def = build_posterior_sd_table(
    joint_trace_def, COMMON_STATS_DEF, 'DEF'
)
```

=====

POSTERIOR CREDIBLE INTERVAL TABLE – LEAGUE EFFECTS

=====

ATT+MID – Posterior League Tax Summary (90% HDI):

League	Stat	Post. Mean	Post. SD	CI 5%	CI 95%
Ligue 1	KP	-0.0244	0.1354	-0.2425	0.1899
Ligue 1	PrgP	-0.2648	0.1581	-0.5095	0.0013
Ligue 1	PrgC	-0.1325	0.1373	-0.3678	0.0883
Ligue 1	SCA	-0.0997	0.1385	-0.3285	0.1237
Bundesliga	KP	-0.1767	0.1685	-0.4551	0.0886
Bundesliga	PrgP	-0.2059	0.1604	-0.4576	0.0625
Bundesliga	PrgC	-0.0251	0.1427	-0.2495	0.2113
Bundesliga	SCA	-0.2283	0.1622	-0.4925	0.0414
Serie A	KP	-0.2018	0.1817	-0.4940	0.0830
Serie A	PrgP	-0.2301	0.1751	-0.5204	0.0542
Serie A	PrgC	-0.0789	0.1492	-0.3356	0.1514
Serie A	SCA	-0.1825	0.1766	-0.4620	0.1129
La Liga	KP	0.1409	0.1684	-0.1287	0.4121
La Liga	PrgP	-0.0041	0.1728	-0.2769	0.2803
La Liga	PrgC	-0.0755	0.1370	-0.2924	0.1499
La Liga	SCA	0.3287	0.1680	0.0565	0.6029

DEF – Posterior League Tax Summary (90% HDI):

League	Stat	Post. Mean	Post. SD	CI 5%	CI 95%
Ligue 1	TklW	0.3403	0.1747	0.0577	0.6243
Ligue 1	Int	0.0173	0.1564	-0.2580	0.2488
Ligue 1	Clr	0.1201	0.1621	-0.1484	0.3781
Ligue 1	Blocks	0.0027	0.1512	-0.2511	0.2463
Bundesliga	TklW	0.1937	0.2049	-0.1480	0.5184
Bundesliga	Int	0.0516	0.1877	-0.2385	0.3614
Bundesliga	Clr	0.0967	0.2039	-0.2565	0.4126
Bundesliga	Blocks	-0.0513	0.1869	-0.3607	0.2565
Serie A	TklW	0.3861	0.2075	0.0792	0.7572
Serie A	Int	-0.0131	0.1766	-0.3049	0.2647
Serie A	Clr	0.3254	0.2196	-0.0181	0.6985
Serie A	Blocks	-0.0680	0.1759	-0.3447	0.2204
La Liga	TklW	0.2599	0.1874	-0.0552	0.5533
La Liga	Int	0.0551	0.1787	-0.2542	0.3324
La Liga	Clr	0.2121	0.1916	-0.0984	0.5161
La Liga	Blocks	0.0977	0.1828	-0.2158	0.3767

In []: # =====

CELL 20b: FIGURES 2 & 3 – POSTERIOR LEAGUE TAX FOREST PLOTS (PUBLICATION)

```

# =====
# Reads from sd_table_att and sd_table_def computed in Cell 20.

plt.rcParams.update(PUB_STYLE)

LEAGUE_COLORS_FOREST = {
    'Ligue 1': '#4878CF',
    'Bundesliga': '#6ACC65',
    'Serie A': '#D65F5F',
    'La Liga': '#B47CC7',
}

LEAGUES = ['Ligue 1', 'Bundesliga', 'Serie A', 'La Liga']

def plot_forest(ax, sd_table, stats_list, title, panel_label):
    n_stats = len(stats_list)
    n_leagues = len(LEAGUES)
    spacing = 1.0
    group_gap = 0.5

    yticks, ylabels = [], []

    for s_idx, stat in enumerate(stats_list):
        base_y = s_idx * (n_leagues * spacing + group_gap)
        for l_idx, league in enumerate(LEAGUES):
            row = sd_table[
                (sd_table['Stat'] == stat) & (sd_table['League'] == league)
            ]
            if row.empty:
                continue
            mean_val = row['Post. Mean'].values[0]
            lo_val = row['CI 5%'].values[0]
            hi_val = row['CI 95%'].values[0]
            y = base_y + l_idx * spacing
            color = LEAGUE_COLORS_FOREST[league]
            excludes_zero = (lo_val > 0) or (hi_val < 0)

            # HDI line
            ax.plot([lo_val, hi_val], [y, y], color=color,
                    linewidth=2.2, alpha=0.8, solid_capstyle='round')
            # Mean dot
            ax.scatter(mean_val, y, color=color, s=55, zorder=5,
                      edgecolors='white', linewidths=0.8)
            # Star for credibly non-zero
            if excludes_zero:
                ax.scatter(hi_val + 0.02, y, marker='*', color=color,
                          s=40, zorder=6)

            yticks.append(y)
            ylabels.append(league)

        # Stat label in left margin
        mid_y = base_y + (n_leagues - 1) * spacing / 2
        ax.text(-0.95, mid_y, stat, ha='right', va='center',
               fontsize=10, fontweight='bold')

    # Separator line between stat groups

```

```

        if s_idx < n_stats - 1:
            sep_y = base_y + n_leagues * spacing
            ax.axhline(sep_y, color='grey', linewidth=0.5, linestyle=':')

    ax.axvline(0, color='black', linewidth=1.2, linestyle='--', alpha=0.7)
    ax.set_yticks(yticks)
    ax.set_yticklabels(ylabels, fontsize=9)
    ax.set_xlabel('Posterior League Tax (log-ratio scale)', fontsize=10)
    ax.set_title(f'({panel_label}) {title}', fontsize=11, fontweight='bold')
    ax.spines['left'].set_visible(False)
    ax.tick_params(left=False)

    # Legend
    handles = [
        mpatches.Patch(color=LEAGUE_COLORS_FOREST[1], label=l, alpha=0.85)
        for l in LEAGUES
    ]
    handles.append(
        plt.scatter([], [], marker='*', color='grey', s=40, label='* 90% HDI excludes zero'
    )
    ax.legend(handles=handles, loc='lower right', fontsize=8, framealpha=0.9)

# Figure 2: ATT+MID Forest Plot
fig, ax = plt.subplots(figsize=(8, 7))
plot_forest(ax, sd_table_att, COMMON_STATS_ATT_MID,
            'ATT+MID League Tax Posteriors', 'A')
fig.suptitle(
    'Figure 2: Posterior League Tax Estimates – ATT+MID\n'
    '90% Highest Density Intervals, relative to Premier League',
    fontsize=11, fontweight='bold', y=1.01
)
fig.text(
    0.5, -0.02,
    'Positive values indicate higher rates expected in PL relative to source league. '
    '* = 90% HDI excludes zero.',
    ha='center', fontsize=9, style='italic'
)
plt.tight_layout()
plt.savefig('figure2_forest_att_mid.pdf', bbox_inches='tight', dpi=300)
plt.savefig('figure2_forest_att_mid.png', bbox_inches='tight', dpi=300)
plt.show()
print("✓ Figure 2 saved")

# Figure 3: DEF Forest Plot
fig, ax = plt.subplots(figsize=(8, 7))
plot_forest(ax, sd_table_def, COMMON_STATS_DEF,
            'DEF League Tax Posteriors', 'B')
fig.suptitle(
    'Figure 3: Posterior League Tax Estimates – DEF\n'
    '90% Highest Density Intervals, relative to Premier League',
    fontsize=11, fontweight='bold', y=1.01
)
fig.text(
    0.5, -0.02,

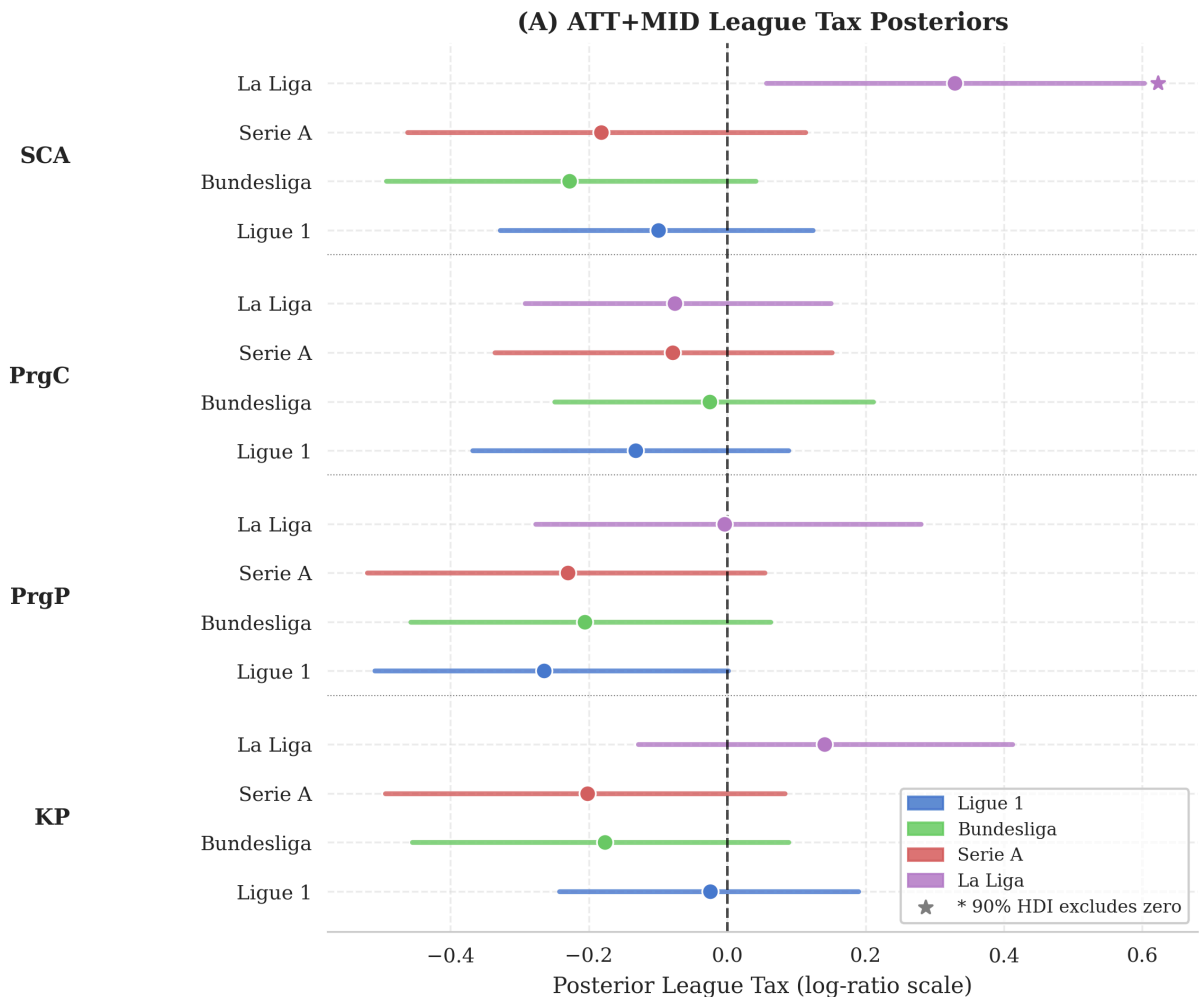
```

```

    'Positive values indicate higher defensive rates expected in PL relative to source league
    '* = 90% HDI excludes zero.',
    ha='center', fontsize=9, style='italic'
)
plt.tight_layout()
plt.savefig('figure3_forest_def.pdf', bbox_inches='tight', dpi=300)
plt.savefig('figure3_forest_def.png', bbox_inches='tight', dpi=300)
plt.show()
print("✓ Figure 3 saved")

```

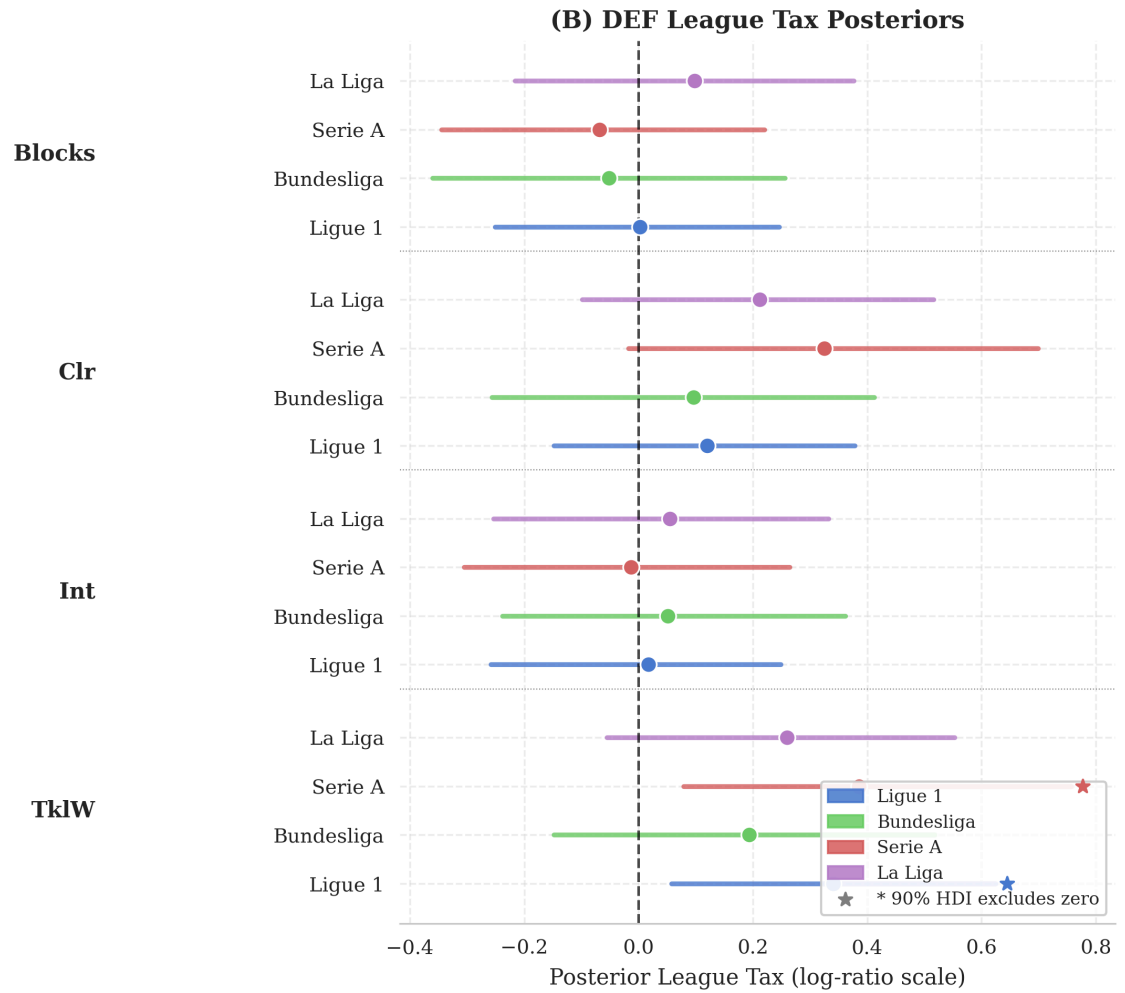
Figure 2: Posterior League Tax Estimates — ATT+MID
90% Highest Density Intervals, relative to Premier League



*Positive values indicate higher rates expected in PL relative to source league. * = 90% HDI excludes zero.*

✓ Figure 2 saved

Figure 3: Posterior League Tax Estimates — DEF
90% Highest Density Intervals, relative to Premier League



Positive values indicate higher defensive rates expected in PL relative to source league. * = 90% HDI excludes zero.

✓ Figure 3 saved